CIMS

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Intro

This is the documentation for the CIMS model.

Part I

install

1 Setup the model

Welcome to the CIMS repository! This guide will help you set up and run the CIMS Python package for energy-economy modelling.

1.1 Prerequisites

Before you begin, ensure you have the following installed:

- **uv**: A Python package and environment manager. Install it with one command — no separate Python installation required.
  - Mac/Linux:

    ```
    curl -LsSf https://astral.sh/uv/install.sh | sh
    ```

  - Windows (PowerShell):

    ```
    powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
    ```

- **Git**: Used to clone the repository. Follow the installation instructions for your operating system.
- **Terminal or Command Prompt**:
  - Windows: Git Bash or PowerShell
  - macOS/Linux: Terminal

1.2 Setup

1.2.1 1. Clone the Repository

Open your terminal and navigate to the directory where you want to work:

```
cd ~/Documents/projects/
```

Clone the repository:

```
git clone https://github.com/EMRG-SFU/cims.git
cd cims
```

1.2.2 2. Install Dependencies

From the `cims` directory, run:

```
uv sync --extra notebooks
```

This will automatically select a compatible Python version, create a virtual environment, and install all dependencies. You only need to do this once, or again when dependencies are updated.

1.2.3 3. Launch Marimo

```
uv run marimo edit
```

Marimo will open in your default browser. From there you can open, run, and edit notebooks.

1.3 Common Workflows

I want to…	Command
Set up or update my environment	`uv sync --extra notebooks`
Launch Marimo	`uv run marimo edit`
Run a Python script	`uv run python my_script.py`
Open a Python shell	`uv run python`
Use a specific Python version	`uv sync --python 3.12 --extra notebooks`

1.4 Subsequent Runs

To re-launch Marimo in future sessions, simply run from the `cims` directory:

```
uv run marimo edit
```

uv will use your existing environment and skip reinstalling dependencies unless something has changed.

1.5 Troubleshooting

- **uv not found after install**: Restart your terminal session so the `uv` command is available on your PATH.
- **Dependency conflicts**: Run `uv sync --extra notebooks` again to bring your environment up to date with the latest `pyproject.toml`.

1.6 Additional Resources

- **Submit Issues**: If you encounter any problems, please submit an issue on our GitHub Issues page.
- **Data Pipeline**: See `data/README.md` for information on the data pipeline and directory structure.

2 Getting started

how to run a scenario

Part II

model

3 Introduction to CIMS

3.1 Background and model objectives

CIMS[1] is an integrated set of economic, energy, and materials models designed to provide information to policy makers on the likely response of firms and households to policies that influence their technology acquisition and technology use decisions.[2] Thus, it is sometimes described as a technology simulation model, which seeks to reflect how people actually behave rather than how they ought to behave.[3] CIMS is able to provide policy makers with information on: (1) the ability of their policies to achieve specific objectives (like greenhouse gas (GHG) emissions reductions), (2) the likely costs of achieving these objectives, and (3) the uncertainties associated with the model's simulations. CIMS is maintained and operated by the Energy and Materials Research Group at the School of Resource and Environmental Management at Simon Fraser University.

CIMS is based on energy flows through a country's economic system. Like its predecessor, the ISTUM family of models, which form the individual sub-components of CIMS, it tracks the flow of energy, beginning with energy services and production processes through to eventual end-use by individual technologies. Unlike the partial equilibrium ISTUM models, which compete technologies against each other to serve pre-specified demands for end-uses, CIMS is nearly a full equilibrium system that allows the practitioner to include price driven energy and goods and services supply and demand equilibrium, energy trade, and some indirect macroeconomic effects, specifically disposable income and savings multiplier effects.

CIMS' focus on energy flows through technologies makes it ideal for modelling policies intended to affect energy efficiency, greenhouse gas emissions, and air quality. Fuel consumption and emissions of all pollutants are technology specific; unless a model operates on a detailed technology basis, as CIMS does, fuel consumption and emission estimates can only be approximated by economic activity.

TODO: Discuss model operational structure (Excel input, Python, Excel results)

TODO: Model licenses (MIT & CC4.0)

TODO: Split repository into code and data

[1]CIMS originally stood for Canadian Integrated Modelling System but, as the model is now being applied to other countries, the acronym is now used as a proper name.
[2]Testing
[3]More testing

3.2 Model Scale and Scope

CIMS covers the entire Canadian economy and can connect to an aggregated representation of the US economy. The Canadian version of CIMS currently includes six provinces and an aggregation of the Atlantic Provinces.[^4] Table 1 illustrates the sectors and regions covered in CIMS.

TODO: Separate out Atlantic provinces (medium priority) and Territories (lower priority)

Table 3.1: : CIMS Sectors for Canada

Sectors	BC	AB	SK	MB	ON	QC	AT
Residential	X	X	X	X	X	X	X
Commercial	X	X	X	X	X	X	X
Transportation Personal	X	X	X	X	X	X	X
Transportation Freight	X	X	X	X	X	X	X
Mining	X		X	X	X	X	X
Coal Mining	X	X	X				X
Natural Gas Extraction	X	X	X	X	X	X	X
Petroleum Crude	X	X	X		X		X
Petroleum Refining	X	X	X		X	X	X
Industrial Minerals	X	X			X	X	X
Iron and Steel					X	X	
Metal Smelting	X			X	X	X	X
Chemical Products	X	X	X		X	X	
Pulp and Paper	X	X			X	X	X

Sectors	BC	AB	SK	MB	ON	QC	AT
Other Manufacturing	X	X	X	X	X	X	X
Waste	X	X	X	X	X	X	X
Agriculture	X	X	X	X	X	X	X

TODO: Include major cities as separate nodes under provinces?

- Would include Toronto, Montreal, and Vancouver only. Use metro region or city only? Is the reason for this to represent city policies?

While the model is simple in operation, it can appear complex because it is technologically explicit and covers the whole economy. CIMS represents a detailed list of technologies (fridges, light bulbs, cars, industrial motors, steel furnaces, buildings, power plants, etc.), and includes their linkages to each other. The model is especially large for the industrial sectors because industry has the largest variety of technologies.

CIMS highlights the interplay of energy supply and demand because energy-related GHG emissions are currently a key policy concern. Thus, the model focuses on the interaction between sectors that use energy (industrial, residential, commercial / institutional, and transportation sectors) and sectors that produce or transform energy (electricity generation, fossil fuel supply, oil refining, natural gas processing, biofuels, and hydrogen). This interaction is shown in Figure 1. A policy that seeks to influence energy supply and demand may also influence the cost of producing goods and services, thus affecting their price and final demand. To reflect these interactions, CIMS includes a goods and services final demand feedback loop, composed of Armington elasticities for traded goods and empirically driven relationships between the traded and non-traded sectors.

3.3 Model Simulation: Six Steps

As a simulation model, CIMS simulates the evolution of technologies acquired to meet energy services, and it uses both engineering and economic information. The engineering information details the unit energy consumption of each energy service technology and explains, using flow models of technologically heterogeneous sectors, how final energy service demands, intermediate products, and energy supply are related. The economic information determines which technologies are preferable in terms of cost minimization. While CIMS is not an optimization model, it can be adapted to produce results like those of an optimization model by calling on certain built-in functions. These functions are known as winner-take-all functions, and they give all new market shares in a given competition to the technology or process with the least cost.

TODO: Review Winner-take-all function and decide if necessary

CIMS is designed to mimic a sector or industry by representing a number of industrial establishments, residential or commercial buildings, and delivery of goods or personal travel within a defined region. The model focuses its analysis on the aggregate energy end-use requirements of that industry or sector. For example, in industry the major end-uses are process heat, motive force, space heat, and lighting. In the pulp and paper sector, the major end-uses are steam heat or mechanical energy to digest wood fibre into pulp, mechanical energy to convey and form pulp into paper, mechanical energy to press moisture out of formed pulp or paper, steam heat to dry pulp or paper, space heat, and lighting. In residential and commercial buildings, the primary end-uses include space heating and cooling, hot water production, refrigeration, and other assorted appliances.

A CIMS simulation comprises six basic steps:

1. The baseline macro-economic drivers, energy supply drivers, and energy service technology characteristics are set for the desired time frame (15 years, 20 years, etc.) in five-year increments.

    a. Technologies are represented in the model in terms of the quantity of energy service they provide. This could be, for example, vehicle kilometers traveled, tonnes of paper, or square metres of floor space heated and cooled. Then, a forecast of growth in energy service demand is used to drive future demand for energy services, usually in five-year increments.

    b. Macro-economic drivers include income, economic output, structural evolution of sectors, interest rates, employment rates, regulations, etc. They account for all energy service demands in the energy demand module.

    c. Energy supply drivers include international commodity prices, domestic market developments, forecasts of costs and availability of marginal energy supplies, and the resulting forecasts of domestic prices.

    d. Energy service technology characteristics include the date a technology or process will be available, technology energy efficiencies and capital / operating costs, firm and household time preferences (discount rates) for discretionary and non-discretionary expenditures (new and retrofit), and other behavioural parameters (perceived risks and consumer surpluses of different technologies, interdependence of certain technology choices that serve different energy services, such as domestic water heating and domestic space heating).

2. The energy demand module (industry, commercial, residential, and transportation) is run for the initial driver values (including energy supply prices).

a. In each future period, a portion of the initial year's stock of technologies is retired. Retirement depends only on age. The remaining technology stocks in each period are subtracted from the forecast energy service demand and this difference determines the amount of new technology stocks in which to invest.

b. Prospective energy service technologies compete to fill the new service demand. The objective of the model is to simulate this competition so that the outcome approximates what would happen in the real world. Here the model relies heavily on market research of past and prospective firm and household behavior. Technology costs depend on recognized and financial costs, but also on identified differences in non-financial preferences (e.g., differences in the quality of lighting from different light bulbs) and risks (one technology is seen as more likely to fail than another). Even the determination of financial costs is not straightforward as time preferences (discount rates) can differ depending on the decision maker (household vs. firm) and the type of decision (non-discretionary vs. discretionary). Consequently, the competition is simulated probabilistically; new stock purchases are distributed among prospective technologies with financial costs being just one of several factors in determining market share.

c. In each period, a similar competition occurs with remaining technology stocks to simulate retrofitting (if desirable and likely from the firm or household's perspective). The same financial and non-financial information is required, except that the capital costs of remaining technology stocks are excluded, having been spent earlier when the remaining technology stock was originally acquired.

3. The demand models then send their requirements for energy to the energy supply models. The supply models send back corresponding energy prices. If the changes in energy prices are over a set threshold value for equilibrium (usually 5%), the demand models are then rerun with the new energy prices. This process is repeated until all energy price changes fall under the threshold value. Figure 2 describes this process:

4. Policy or market driven energy price changes may change the lifecycle cost of some energy services, thereby changing the end-use price for these services. If the change in lifecycle cost from the business-as-usual case is beyond a pre-set threshold price elasticities for these services are applied, thereby changing the demand for these services. This new demand is sent back to the supply and demand models for re-calculation.

5. The model reiterates step 2, 3 and 4 until all variables stabilize. It then proceeds to the next time period. Steps 1 through 5 are repeated for all time periods.

6. Output: Output is provided for total capital stock, new capital stock, fuel use, service use, emissions, levelized costs by technology, and energy service prices.

   a. **Sub-model Outputs**

   i. Annualized lifecycle costs by sector by year by technology based on behavioural discount rates (20-80%) (See Table 5).
   ii. Fuel use and emissions by sector by year by technology.
   iii. Service and auxiliary use by sector by year by technology.
   iv. Total stock use by sector by year by technology.
   v. New stock use by sector by year by technology.
   vi. Retrofits by sector by year by technology.
   vii. Technology cost changes due to declining capital cost function

b. **Supply and Demand Feedback Outputs**
   i. Annualized lifecycle cost per unit produced for all sectors by year based on financial cost of capital for both the BAU and Policy case; the model operator sets the cost of capital (e.g. 7.5 or 10%)
   ii. BAU and Policy final demand for each sector's product or service by year.
   iii. BAU and Policy disaggregated energy demand for all sectors by year.
   iv. Changes in disaggregated capital, operations and maintenance, and energy costs by sector by year.
   v. Changes in GHGs and selected criteria air contaminants (CACs), energy and service use by technology, sorted by technology competition, for BAU and Policy by sector by year.
   vi. Changes in net exports of key energy commodities from BAU to Policy
   vii. Changes in energy prices by year
   viii. Changes in the financial cost of production split into the cost of production and demand effects for industry and manufacturing
   ix. Changes consumers' surplus for personal transportation and residences
   x. Changes in value-added for industry and manufacturing

TODO: Model currently equilibrates only through prices. Could we also find quantity equilibrium for cap system?

- Would need special market (?) for emissions cap that doesn't need specific service requests
  - Include emissions from all capped sectors in market
  - Iterate carbon price until emissions match cap

  - Use shadow price like for DAC? Iterate between max quantity cap and min tax price at each node
  - can we generalise this for DAC, trading cap, any quantity max
- Is this worth the effort or just iterate carbon price manually?

4 Technology Competition and Demand Feedbacks

TODO: If setting market shares after the base year, need to use lifetime = 1, otherwise stock is not calculated correctly.

CIMS operating equations and their subcomponents are presented below in the order established in Section 2: sub-sector technology choice, energy supply and demand equilibrium, and finally goods and services demand equilibrium.

4.1 Sub-model Technology Choice Dynamics and Equations

Sub-model technology choice occurs according to the following order:

- Demand Assessment
- Retirement
- Assessment of capital stock availability
- Retrofit & new capital stock competition to meet demand not supplied by existing capital stock

4.1.1 Demand Assessment

All of the CIMS energy demand models (industry, transportation, commercial / institutional and residential) start with an exogenously determined demand for physical goods and services in each run period. These demands are mostly taken from Natural Resources Canada forecasts. The energy supply models may either use exogenously determined forecasts or they can be run using energy demand calculated from the demand model plus net exports.

4.1.1.1 Request-Provide demand

4.1.1.2 Distributed supply

4.1.2 Capital Stock Retirement

CIMS begins each simulation with an existing base stock of technologies. A linear function is used to retire the existing stock of unknown age already in use. Each year a constant amount of the base stock is retired until there is no more base stock remaining.

TODO: Sunset tech stock that falls below a threshold.

Need to develop rules around when to apply (for example, only after tech reaches some market share should this function be allowed)

Equation : Existing stock retirement

$$BS_t = BS_0 \left(1 - \frac{T_t - T_0}{L}\right)$$

Where:

BS_t is the base stock at time t

BS_0 is the initial base stock

T_t is the current simulated year

T_0 is the initial year

L is the lifespan of the technology

Retirement of new capital stock purchased during CIMS simulations is calculated using a different methodology than existing base stock because the vintage of the base stock is of uncertain and varied vintage, while the vintage of new stock is known.

Equation : New stock retirement

$$NS_t = \frac{NS_p}{1 + e^{\frac{11.513}{L}((T_t - T_p) - L)}}$$

Where:

NS_t is the new stock not yet retired at time t

NS_p is the new stock acquired at purchase time p

T_t is the current simulated year

T_p is the year at purchase time p

L is the lifespan of the technology

The intercept value of 11.513 comes from ISTUM for average technology retirement rates

This function uses an s-shaped, logistic curve to simulate the retirement of devices. Figure 5 shows an example of the retirement function of a technology with a two-year lifespan. Half of the stock is retired before the average lifespan of the technology due to early replacement and failure and half is retired afterward to simulate technologies that last longer than the average.

TODO: Add note about intercept uses in different sectors (i.e., 11.513 for products, but more like 3 for houses)

: New stock retirement function

The total stock of a technology in any given year t is the sum of the unretired base stock and the unretired new stock from each previous year.

Equation : Total stock

$$Stock_t = BS_t + \sum_{p=0}^{t} NS_t$$

The *Stock* at time t is the sum of unretired new stock (*NS*) from all previous years added to the remaining base stock (*BS*).

4.1.3 Retrofit and New Stock Competition

While retrofitting occurs before new stock competition, the retrofitting equations are a subset of the new stock equations and are therefore presented after them. The market shares for individual technologies are determined using the equations below.

4.1.3.1 New Stock Competition

TODO: Check if Winner Take All is included in Python code (need this?)

TODO: Still want market share based competition for VREs?

serv_req_ms – ms – branch

serv_req_ms – amount - branch

Equation : Market share

$$MS_{kt} = \frac{LCC_{kt}^{-v}}{\sum_{j=1}^{J} LCC_{jt}^{-v}}$$

Where:

MS_{kt} is the market share of technology k for new equipment stocks at time t

LCC_{kt} is the annual lifecycle cost of technology k at time t

v is the variance parameter representing cost homogeneity

j technologies compete to provide the same service as k

The MSkt function is a logistic curve whose slope is determined by a variance parameter, v. A high value for v, such as 100, means that the technology with the lowest lifecycle-cost captures almost all of the new equipment stocks, as would occur with a linear programming model. An extremely low value for v, such as 1, means that new equipment market shares are distributed almost evenly between all competing technologies, even if their lifecycle-costs differ significantly. Thus, v represents sensitivity of the technology competition to relative lifecycle-costs.

For nodes that compete technologies that come in two different sizes, market shares are determined separately for each size. After the market shares for each size have been allocated, they are averaged using the Size and Capacity Utilization (SCU) weights:

Equation : Lifecycle cost

$$LCC_{kt} = CC_{kt} + OM_k + SC_{kt}$$

Where:

LCC_{kt} is the annual lifecycle cost of technology k at time t

CC_{kt} is the capital cost of technology k at time t

OM_k is the operating costs of technology k

SC_{kt} is the service cost of technology k at time t

Equation : Annual capital cost

$$CC_t = \max\{DCC_t, CC_{2000} * DCC_{limit}\}$$

Where:

CC_t is the annual capital cost of a technology at time t

DCC_t is the declining capital cost at time t

CC_{2000} is the capital cost of a technology in the year 2000 (before applying the declining capital cost function)

DCC_{limit} is the lowest value to which a technology's capital cost can be reduced by the DCC function (as a percentage of the capital cost in year 2000)

See the section below for the declining capital cost functions.

Equation : Capital cost in year 2000

$$CC_{2000} = \left(\frac{CC_{overnight} + FIC + DIC_t}{Output}\right) * CRF$$

Where:

CC_{2000} is the capital cost of a technology in the year 2000 (before applying the declining capital cost function)

$CC_{overnight}$ is the overnight capital cost

UIC_{fixed} is the upfront fixed intangible cost meant to represent non-financial costs for investment

$UIC_{declining,t}$ is the upfront intangible cost that can decline over time

Output is the material output of a technology

CRF is the capital recovery factor

Equation : Capital recovery factor

$$CRF = \frac{r}{1 - (1 + r)^{-N}}$$

Where:

CRF is the capital recovery factor of a technology

r is the financial discount rate

N is the lifespan of a technology

Equation : Service cost

$$SC_{kt} = \sum P_{jt} * R_{kjt}$$

Where:

SCkt is the service cost of technology k at time t

Pjt is the price of service type j at time t (can be either fuel price or the weighted-average lifecycle cost of a service based on market shares of technologies competing to provide that service)

R_{kjt} is the service requested for technology k of service type j at time t

Note that market shares used to calculate the weighted-average LCC of a service are from the previous iteration of the demand-supply equilibrium process (i.e. the final market shares from the previous year if calculating the first iteration or the previous demand-supply iteration if equilibrium prices have not yet been found).

4.1.3.2 Retrofit

TODO: Consider revising retrofit assessment

Current: OPEX vs competing techs total LCC -> are there cheaper alternatives to replace incumbent?

gTech: OPEX vs node Service cost -> am I losing money operating this tech given the service revenue?

The retrofit option allows functioning equipment (or parent technologies) to be converted to a different type of equipment (retrofit technologies) before its retirement date. For example, certain boilers can be retrofitted into cogenerators before they are retired. The retrofit calculations process is roughly the same as that for new stock calculations, with a couple of changes.

After existing stock is retired, CIMS checks to see if the retrofit switch is on and if any retrofit options are available. If so, CIMS treats the initial purchasing price of the parent technology as a sunk cost and uses the operating and maintenance (O&M), energy, service and emissions costs as the only costs of the parent technology. The costs of the retrofit options are then calculated as the sum of O&M, energy, service and emissions costs plus the annualized retrofit capital cost less any tax or other credit. CIMS then determines, using a similar competition calculation to those already described, how many parent technologies will be retrofitted.

The competition between the parent technology and its retrofit options is based on a standardized Weibull distribution with a shape parameter = 2, location parameter = 0 and variance = 1. This is slightly different from the new stock competition which uses an inverse power function. The analyst can change the variance in the Economic Data File: Economic Description, and the mean reflects the annualized costs of the parent technology and its retrofit options. CIMS randomly chooses 100 points from this distribution for each parent and retrofit technology. The cost of the technology is multiplied by this Weibull value and the variance (see Table 1.1 from the ISTUM manual for a description of what the variance choice entails). Then a matrix is formed (number of technologies by 100 columns or NTECH;100) of technology costs. CIMS reads the columns to see which of the technologies in any one competition is cheapest and then tallies the winners. The ratio of wins/100 for each parent or retrofit technology determines its share of the stock on hand at the beginning of the simulation year or the "remembered" stock. Components in the output APL file store the information so that, in future simulation years, CIMS knows the degree of retrofit in this original stock. CIMS assumes that the degree of retrofit experienced in the simulated year occurred in each previous, unsimulated year.

The analyst can place some bounds on this retrofit option. A maximum (R-MAX) or minimum (R-MIN) market share can be established such that only a certain amount of the old stock could (maximum) or must (minimum) be retrofitted in any year. In an unconstrained market, maximum should be set to 1 and minimum to 0. Because Monte Carlo simulations determine the level of retrofit, the typical retrofit procedure leads to less than the maximum and more than the minimum. For example, if we permit the lead technology, say oil boilers, to retrofit to gas boilers with the R-MAX set to 75%, 25% of the parent stock is removed from the retrofit competition and the remainder undergo the retrofit competitions. Because it is unlikely that Monte Carlo simulations will result in all the remaining stock being retrofit, the total retrofit will usually be less than 75%. On the other hand, if R-MIN were set to 40%, then 40% of the parent stock are forced to retrofit and the remainder allowed to undergo the competition. Because it is likely that Monte Carlo will select at least some of the remaining stock to be retrofitted, the total retrofit will always be greater than 40%.

All the parameters discussed above may be changed in Economic Data: Economic Description File.

4.1.4 Modifications to the technology choice equations

Several functions have been added to the competition process to reflect both real world conditions and policy concerns. These currently include functions to:

- Reduce capital costs as total production increases to reflect learning by doing and efficiencies of scale;
- Reduce intangible costs as new market share climbs to reflect reduced perceived risk (i.e. the neighbour effect); and
- Eliminate obsolete but still functioning technologies.

4.1.4.1 Declining Capital Cost

The declining capital cost function allows the capital costs of technologies to decline as a result of learning by doing and improving economies of scale. In CIMS, capital costs decline as a result of the cumulative production of a technology across Canada and an exogenous rate of decline which represents production in other countries.

Equation : Declining capital cost

$$DCC_t = GCC_t * \left[\left(\sum_{1}^{p} \frac{CumulNS_{2000,p} + \sum_{j=2005}^{j=t-5} NS_{jp}}{CumulNS_{2000,p}} \right)^{\log_2(PR)} \right]$$

Where:

DCC_t is the declining capital cost of a technology at time t

GCC_t is the capital cost of a technology adjusted for cumulative stock in other countries

NS_{jp} is the new stock of a technology added from 2005 to the previous time period in each province, p

$CumulNS_{2000,p}$ is the cumulative new stock of a technology for all years up to and including the year 2000 in each province, p (exogenous)

PR is the progress ratio

Increased stock in other countries reduces the capital costs of a technology in CIMS. Therefore, even if there is no change in stock of a technology in Canada, its capital costs may still decline. The rate at which the capital costs decline as a result of global stock change is set exogenously and is equivalent to the autonomous energy efficiency index (AEEI).

Equation : Global capital cost reduction

$$GCC_t = GCC_{t-5} * (1 - AEEI)^5$$

Where:

GCC_t is the capital costs of a technology after it has been adjusted for cumulative stock in other countries

AEEI is the exogenous rate of decline of a technology's capital costs

An example of the calculations for the declining capital cost function are illustrated below. The capital cost of the technology is $100 in year 2000, its PR is 0.8 (set in column J), and its exogenous rate of decline (set in column U) is 1%:

Additional Issues and Problems with the DCC function

- To use the DCC function it is necessary to activate a global switch in the CIMS Simulation Creation Centre. Look under 'Select Global Parameters' – 'Technology Parameters'. However, please note that this switch only affects the policy simulation runs and not the BAU run. To turn off the DCC function for BAU runs it is necessary to reset the parameters in the Economic File.

- In order to prevent capital costs from declining indefinitely, an analyst may specify the percentage of the initial capital costs below which capital costs may not decline. The percentage is set in column V of the Economic File, and it must be multiplied by 10,000 (ie 5% * 10,000 = 500).
- We don't currently track learning spillover between technologies. Research has shown that investment in one technology (e.g., hybrid cars) can lower costs of another technology (e.g., battery cars).

Summary of DCC parameters in the Economic File

Parameter description Name Column Column name*

Progress ratio PR J Asymptote

Total global production of the technology up to the year 2000 No K Maturity

The exogenous rate of decline of a technology's capital costs ERD U Years Remaining

Percentage of the initial capital costs below which capital costs may not decline (x10,000) V Year Available

* the column names in the Economic File are no longer relevant and should be ignored

4.1.4.2 Declining Intangible Cost Function

(In ISTTMPBG.ws forward).

The declining intangible cost function applies to both capital and operating costs, and allows intangible costs to fall as a results of falling perceived risk as market exposure increases. The following general formula applies.

(Equation 20)

where:

- It is the intangible cost in time t
- I0 is the initial intangible cost, which is column N of the economic file for capital and column R for the operating costs.
- A is a constant in column O of the economic file for capital, and column S for operating costs.
- K is a constant in column P of the economic file for capital, and column T for operating costs.
- NMS new market share

• NMS(t – 5) is the percent of the new market share in from the previous run. This value should be between 0 and 1 (i.e., between 0% and 100%), as opposed to between 0 and 100. NMS is the share of the technology from all other technologies that provide the same service.

4.1.4.3 Technology Die-Out Function

(In ISTTMPBT.ws forward).

The die-out function is designed to eliminate technologies from the market when their stock falls below a proportion of their market share in the base year.

The die-out function will be incorporated into the market share for new capital stock equation. The old methodology for calculating market share (equation 4) is given below:

New methodology:

(Equation 21)

where:

- MSkt = the market share for technology k in time t
- LCCkt = is the lifecycle cost of technology k in time t
- v = the variance parameter representing cost homogeneity
- J = the number of technologies competing to provide the same service as k
- n = the threshold factor where a technology is eliminated from the market. For n = 0.5, if a technology falls to 50% of its market share in 2000, it will be eliminated from the market for new capital stock. (n applies to all technologies and cannot be set individually for different technologies)
- f = the base year. The base year can be set to any year. (f applies to all technologies and cannot be set individually for different technologies)

The new methodology for calculating the market share of a technology will eliminate a technology if its market share falls below its threshold (n x MSkf). Following a technology's elimination from the market, it should not be allowed to compete in the following year.

The analyst is able to set the value for n and f. However, the default setting for n should be 50% or 0.5, and the default setting for f should be 2000.

4.1.4.4 Expectations Functions

The expectations functions have been introduced to simulate situations when businesses and consumers have foresight into their future emissions costs. The functions calculate the expected emissions costs of individual technologies in CIMS, in order to account for technologies with different lifespans and discount rates.

CIMS has three options for generating expectations: 1) myopic, where people do not have foresight into their future emissions costs; 2) perfect foresight based on the average emissions costs of a technology; and 3) perfect foresight based on the discounted value of emissions costs of a technology.

Myopic

This option is identical to CIMS's previous methodology for calculating emissions costs. The emissions charge used to calculate emissions costs over a technology's lifespan is the emissions charge set in CIMS's CIMS Simulation Creation Centre / Select Global Conditions / Emissions.

E(Emissions Charge)kt = Emissions charget

The expected emissions charge used to calculate the lifecycle costs of technology k, in the year of its purchase (year t), is the emissions charge set in the Simulation Creation centre.

Average Expectations

The second option generates expectations of future emissions costs based on the average emissions charge over the lifespan of a technology.

(Equation 22)

where:

EmissionsChargen is the emissions charge occurs in year n, and set in CIMS's Simulation Creation Centre;

base is the year technology k is purchased;

N is the lifespan of technology k.

Discounted Expectations

The third option generates expectations of future emissions costs based on the present value of emissions charges. After the present value of emissions charges has been calculated, it is annualized using the capital recovery factor. The present value must be annualized in order to obtain an annual value of emissions charges, which is required in CIMS's simulation.

(Equation 23)

where:

EmissionsChargen is the emissions charge occurs in year n, and set in CIMS's Simulation Creation Centre;

base is the year technology k is purchased;

N is the lifespan of technology k.

rk is the revealed discount rate of technology k, and it is set in the Techno Economic file.

Additions:

One of the challenges in introducing expectations into CIMS is that emissions charges can only be set up to 2035. To overcome this challenge, emissions charges are assumed to remain the same as in 2035 for years beyond 2035.

(Equation 24)

Switch for Allocated Costs

In Column L of the Economic File, an analyst may specify a value to reduce the capital costs of a technology. If the reduction is a subsidy, it should not represent an actual reduction (or increase) in the capital costs of the technology; and it should not reduce (or increase) the measure of capital costs in the BAUP or CONP files. However, if the value in column L represents a real cost (not a subsidy), it should be included in the measure of capital costs in the BAUP or CONP files. ISTUM enables an analyst to use alternative definitions for the value in column L. If the value is a "real cost", then set is value to "0" in ISTUM; if the value is a transfer (or subsidy), set the value to "1".

4.1.5 Energy Supply and Demand

In CIMS's integrated energy supply and demand system, the demands of transportation, industry, residences, and the commercial & institutional sector directly drive the activities of electricity production, refining, coal mining and natural gas and crude oil extraction. Inter-regional transfers as well as net exports are added, as demand for Canada's energy supply sector includes US demand. The importance of domestic demand varies by energy commodity. Most electricity is produced for Canadian consumers, more than half of natural gas produced is shipped to the US, and there is a heavy trade in crude oil and refined petroleum products. Half our domestic crude oil production is shipped to the US, while we import from both OPEC and non-OPEC suppliers.

Each of the energy markets in CIMS has the option of fixed or dynamic pricing, and they are described below:

- Seven regional electricity markets are included, each with endogenous internal pricing based on the cost of production. The amount of electricity produced is the sum of endogenous domestic demand and net extra-regional exports, which may be adjusted with using an export own-price elasticity.

- One national natural gas market is included, with dynamic pricing using a supply curve based on a combination of Canadian and US market feedbacks. The amount of gas produced is the sum of endogenous domestic demand and net exports. Net exports may be adjusted to reflect changes in the domestic cost of production using an export price elasticity.
- One national crude oil market, based on an exogenously specified world price of crude oil. The amount of crude oil produced is the sum of endogenous domestic demand and net exports. Net exports may be adjusted to reflect changes in the domestic cost of production using an export elasticity. The world crude oil price is assumed to be unresponsive to Canadian demand.
- Six refining regions divide their production amongst seven regions and net exports. All six can use endogenous internal pricing based on the cost of production. The amount of refined petroleum products produced is the sum of domestic regional demand and net extra-regional exports.
- Four coal producing regions divide their thermal coal production amongst endogenous domestic regional demand and exogenous exports (i.e. BC exports 100% of its production while Alberta exports about 30% of its production). Due to the differences between the ways each non-producing region and sector gets its coal, these regions are assumed to buy coal at a fixed market price or according to a local supply curve – for example, some electricity plants in Alberta and Saskatchewan are sited at coalfields and are the sole customers. British Columbia, Alberta, Saskatchewan and the Atlantic provinces all buy their coal at a fixed market price.

CIMS divides energy flow in the economy into demand for energy end-use services, intermediate processing, and primary supply (Figure 7). It begins with an initial demand for four general end-use energy commodities: electricity, natural gas, coal, and refined petroleum products. The electricity needs of the domestic energy demand models (transportation, residential, commercial and industry) plus net electricity transfers (inter-regional transfers and exports) are used to drive electricity generation. The refined petroleum products sector is driven by the net demand from the energy demand models, electricity, inter-regional transfers, and net exports. The primary energy supply models, crude extraction, coal mining and natural gas extraction and transmission are driven by the net demand for their products from the final demand sectors, electricity and refined petroleum.

Figure 3. CIMS'Energy Supply and Demand Flow Model

The demands for electricity by petroleum refining and the primary energy supply models, for refined petroleum products by the primary energy supply models, and for primary energy supply by the other primary energy supply models, would ideally be endogenous to the system. However, making these demands endogenous would have required another set of feedback loops for each of electricity, refined petroleum products, natural gas, and coal and crude oil production. The effect of each of these potential loops was tested manually and they did not have a significant impact on the net effects of either a large GHG tax or change in energy price. We decided not to include these loops in order to simplify the model. A system of interior loops, or one that solves the demand for electricity, natural gas, crude oil products and coal simultaneously may be added to CIMS in the future.

Canada trades a significant amount of energy with the United States and, to a lesser extent, with both OPEC and non-OPEC suppliers. Given the possible effect of Canadian energy policy on the price of producing these commodities, and hence their marketability to the US, electricity, crude oil and natural gas export feedbacks are included in CIMS using price elasticities calculated for the Canadian Federal Department of Finance (Table 5).

Table 5. Price Elasticities for Energy Exports, Wirjanto (1999)

Energy Commodity Long-Run Own-price Elasticity for Export Demand

Electric power -0.54

Natural gas -0.67

Coal -0.51

Gasoline and fuel oil -0.92

Other petroleum products -0.82

These elasticities follow the Armington specification, whereby the demand for an imported good rises as its price falls relative to the price of its domestic substitute. At all times, consumers buy at least some of the domestic good and some of the imported good, even when the price difference is large (Iorwerth et al. 2000, Wirjanto 1999). Please see Bataille (2005) for further elaboration. The Armington specification assumes that goods produced by different countries are not perfect substitutes. The energy trade elasticities are represented in Equation 20, which applies the elasticity of the ratio of import volumes to domestic production in relation to a change in the relative price of imports to domestic prices.

(Equation 25)

Where:

FLCCt = Current financial lifecycle cost for producing the energy commodity

FLCCt-1 = Previous financial lifecycle cost for producing the energy commodity

COP = Cost of production covered by the energy producing sub-model, as a % of total cost

= Price demand elasticity for the energy commodity

IT= Initial Trade volume of the energy commodity

Adjprod = Additive adjustment to production of the energy commodity

Western Canadian exports of crude oil are separated from eastern Canadian imports, as there are currently no large cross-country shipments. While electricity trade is currently a very small portion of our energy trade, electricity import substitution elasticities are included to cover the possibility of a policy driven increase or decrease in electricity exports to the US. All these elasticities operate on the business-as-usual (BAU) export amounts. These elasticities can be single values or they may change over time as specified by the operator of the model.

4.1.6 Goods and Serviced Demand Feedbacks

The economy is commonly divided into: households, who provide labour and financial capital and who consume goods and services; firms, who consume capital and labour and provide goods and services; government, which consumes goods and services, provides public goods and regulates the market; and finally foreign producers and consumers of goods, services and capital (Dornbusch et al. 1989). It is common to assume that Canada is a price taker for capital (i.e., the interest rate is set at the world price of capital).

> Canada is assumed to be a small open economy . . . The world interest rate is assumed to be the US prime rate, and foreign savings are assumed to be available elastically. This means that the foreign sector automatically clears any surplus or deficit in the domestic savings market each period, so the domestic interest rate is effectively exogenous. (McKitrick 1998) p.552

I maintained the same assumption as McKitrick, that capital is infinitely available at the world market price. This assumption left the circular flow of goods and services between producers (domestic and foreign firms and government, through "public goods" provided via hospitals, schools, and other institutional buildings) and consumers (households, government, and domestic and foreign firms). ISTUM is a model of the physical production side of the economy – the demand side of the economy is mostly missing. To complete the circle between producers and consumers, the system required the addition of demand feedbacks.

The following suite of options face the consumer or producer when the price of a good or input rises, modelled after the consumer's utility map in CASGEM (Iorwerth et al. 2000):

- Substitute an import.
- Buy something else.
- Substitute current consumption for savings.
- Work less and enjoy more leisure, which is the same as reducing output.

These options were used as the primary guide for adding demand feedbacks to the ISTUM sub-models in the construction of CIMS.

4.1.6.1 Demand Feedbacks for Manufactured Products

Most Canadian industrial products are highly traded; their sensitivity to foreign competition is of prime importance. To reflect the sensitivity to foreign competition, CIMS employs the same Armington specification for price elasticities as used for energy trade, whereby the demand for a foreign good rises as its price falls relative to the price of a domestically produced substitute. Using the Armington specification, consumers always buy at least some of the domestic good and some of the imported good. The price elasticities are applied with the general formula in Equation 21.

(Equation 26)

Where:

FLCCt = Current financial lifecycle cost for the product

FLCCt-1 = Previous financial lifecycle cost for the product

COP = Cost of production covered by the sub-model that produces the product, as a % of total cost

= Price demand elasticity for the product

Multiplier = Value by which demand for the product is multiplied

To accommodate the differences between products in the industry sub-models, especially Industrial Minerals, Other Manufacturing, Chemical Products, and Metal Smelting, demand feedbacks are applied to these models at the level where production for individual products is determined. Table 8 provides the elasticities used for sectors and their products. See Appendix E of Christopher Bataille's Ph.D dissertation for a full breakdown of their sources and how they were mapped to CIMS' sub-models and their sub-products.

Table 6 Price elasticities used from Wirjanto (1999)

4.1.6.2 Demand Feedbacks for Commercial and Residential

There is no import substitution for non-traded goods and services such as residential and commercial floor space. CIMS uses a log-linear econometric relationship to estimate the relationship between manufacturing activity – used dynamically as valued added but proxied using the value of shipments – and residential / commercial activity – measured by the value of building permits. The resulting relationship is used to allow commercial and residential activity to respond to changes in manufacturing activity.

(Equation 27)

Where:

BPc,r = natural log of activity (value of building permits) in the commercial or residential sectors in a given year

IVA = natural log of industrial value-added in a given year

e = error term

m = coefficient of relationship (elasticity)

b = intercept

This equation was estimated, using SHAZAM v.9, by running ordinary least squares regressions between manufacturers' shipments and residential building permits between 1961 and 1991 and manufacturers' shipments and commercial, institutional, and governmental building permits between 1961 and 1991. All values were deflated using a standard consumer price index. We tested various permutations of the regressions, including lagging and the use of population and year as an independent variable. Table 9 provides the parameter estimates.

Table 7. Parameter Estimates Used to Provide Demand Feedbacks for the Residential and Commercial/Institutional Sectors

The process for calculating manufacturing valued-added, the driver for residential and commercial activity in CIMS, is complex. The sub-model's outputs are designed to capture changes in energy, new capital and labour expenditures engendered by policy. These costs are a sub-component of the total cost of production, which also includes materials and supplies, as well as capital investment, labour and repairs not related to energy using equipment.

Value-added is defined as the total value of shipments and other revenue, minus the cost of intermediate inputs (energy, materials, and supplies) (Lipsey et al. 1998, p.175). Returns to capital include both amortization and profits. Using values for wages, energy expenditures, costs of material, total value of shipments and capital expenditures from Statistics Canada, we found that while the ISTUM sub-models do not capture expenditures for materials or capital unrelated to energy consumption, they capture most capital expenditures in the sectors for which sub-models exist. Supplemental to this, the demand feedback function of CIMS, when faced with a reduction in demand, removes first new growth, then the necessity for replacing retired technology stock, and finally useful stock that is no longer required. Given that most sectors in CIMS grow in most time periods, a simplifying assumption was made that all the productive investment being removed was new or replacement investment, not useful stock. We then compared CIMS' new investment in 2000 with figures from Statistics Canada, and derived a set of multipliers to scale up CIMS' investment figures to reflect the ratio of coverage (the multipliers affect the demand for all stock, not just the portion of the cost of production covered by ISTUM). Finally, we added normal profits as a percentage of revenue, having ascertained that normal profits are included in value-added and that these are not normally included in CIMS. These were calculated as a percentage of total costs from Statistics Canada's "Financial and taxation statistics for enterprises, 2000, 61-219".

4.1.6.3 Demand Feedbacks for Freight Transportation

Freight transportation is important for energy and emissions policy because of the large amount of energy consumed and the amount of GHGs and criteria air contaminants produced. Activity in this sector is directly related to the amount of activity in the rest of the economy. To provide feedback relating freight and economic activity, a similar relationship to that established between manufacturing value-added and residential and commercial activity was incorporated into CIMS. It proved difficult to find an equivalent statistic for freight activity to manufacturing activity as used for the commercial and residential sectors, so instead of an estimated relationship a rule-of-thumb value was temporarily used instead (0.95 – a 1% fall in manufacturing value-added causes a 0.95% fall in freight activity). This relationship allows changes in manufacturing value-added to adjust freight transportation activity. This relationship will be estimated once statistics are available.

4.1.6.4 Demand Feedbacks for Personal Transportation

Demand feedbacks for personal transportation operate in a similar fashion to the demand feedbacks for manufacturing and industry, except that in the place of an Armington price elasticity an elasticity for personal kilometers traveled (pkt) in response to the cost of travel is used instead (price elasticity =-0.02) (Michaelis and Davidson 1996). Transportation is also a bit different from the other sub-models in that some demand adjustments occur within the sub-models, specifically the choice between urban modes of transport (single occupancy vehicles, high occupancy vehicles, transit and walking and cycling). The upshot is that much of the structural change possible in transportation is directly included in the sub-sector models.

4.1.6.5 Limits to the Demand Adjustments

Experimentation with early versions of CIMS showed that the demand adjustments had to be regarded critically. Elasticities are by nature log-linear measurements, while demand responses are very often non-linear. It may be hypothesized that there will always be a demand for some commodities, no matter the price. A function for limiting their effects was installed; 50% was chosen as the maximum demand reduction as a rule-of-thumb for the maximum extent to which CIMS' substitution elasticities were still applicable. This demand limiting adjustment is somewhat simplistic, and in the future some form of function that applies limits to demand adjustments in a progressive way will be installed.

4.1.7 Building the tree

Need a section in here about best practice for designing nodes.

For example, should request services directly at node and also have tech competition.

5 Model Sector Descriptions

5.1 Fuel Pricing

TODO: Change from Fuel to Energy (or Supply) throughout all documentation, model, and code?

5.1.1 Primary Energy

TODO: How define a energies in CIMS to align with other reporting frameworks, like RESD?

5.1.1.1 Fixed Price

5.1.1.2 Cost Curve – Annual

5.1.1.3 Cost Curve – Cumulative

5.1.2 Secondary Energy

TODO: How define a fuel in CIMS

Calibration:

1. Set Financial lifecycle cost to 1
2. Set Price Multiplier to expected price (including sector markup)
3. Calibrate demand sector
4. Calibrate fuel supply sector to achieve expected production cost at node one below final
5. Remove Financial lifecycle cost
6. Reset Price Multiplier to only include sector markup

5.1.3 Carbon pricing

TODO: Have additive carbon prices to allow for overlapping policies.

For instance, carbon tax + shadow price for cap.

5.2 Upstream fossil fuel extraction

The petroleum crude industry in Canada includes:

- A conventional crude oil production system, which comprises all facilities and associated infrastructure used to extract, treat, and transport light, medium, and heavy crude oils to market.
- An unconventional crude oil system, which comprises all facilities used to extract crude bitumen from oil sands and subsequently upgrade them to synthetic crude oil. It is disaggregated into in-situ bitumen recovery and oil sands mining/upgrading.

5.2.1 Conventional crude oil

The production, treatment and transmission segment of the conventional crude oil system may comprise the following infrastructure:

- Oil wells which produce at least one barrel of crude petroleum oil to each 100,000 cubic feet of natural gas.
- Oil transmission pipelines or tanker vehicles to transport the well effluent to a battery or a cleaning plant.
- Batteries where effluent from wells is separated into its constituent phases for metering and appropriate disposition.
- Oil storage tankers.
- Oil transmission pipelines or tanker vehicles to move crude oil from producers to market.

5.2.2 Oil sands mining/upgrading

Oil sands mining/upgrading usually comprises the following steps:

- Surface mining: Since the 1920's, open pit mining has been central to oil sands development. Mine equipment from the early years was scaled up significantly when large commercial operations started to come on line. In the past, draglines and bucket-wheel reclaimers were used to mine oil sands. This practice has largely been replaced by shovel-and-truck mining, which gives greater flexibility. Run-of-mine oil sands, dug directly from the open pit faces, is hauled to in-pit crushing stations, and is then slurried and pumped to the bitumen extraction plants.
- Bitumen extraction: The oil in oil sand is called bitumen, a complex hydrocarbon made up of a long chain of molecules. Bitumen is usually separated from oil sands using a warm-water extraction process. The mixing that takes place during slurry transport from the mine to the plant is sufficient to begin the separation process, which recovers over 90% of the bitumen resource. The resulting bitumen froth then forms the feedstock for creating synthetic crude oil.
- Raw bitumen upgrading: In order for bitumen to be processed in refineries, the complex hydrocarbon chain in bitumen must be broken up and reorganized, which means removing some carbon while adding additional hydrogen to make more valuable hydrocarbon products by upgrading. This is done using four main processes: coking removes carbon and breaks large bitumen molecules into smaller parts, distillation sorts mixtures of hydrocarbon molecules into their components, catalytic conversions help transform hydrocarbons into more valuable forms and hydrotreating is used to help remove sulphur and nitrogen and add hydrogen to molecules. Further treatment and blending produces light low-sulphur (sweet) synthetic crude that is shipped by pipeline to refineries.

5.2.3 In-situ Bitumen

About 80% of the oil sands in Alberta are buried too deep below the surface for open pit mining. This oil must be recovered by in situ techniques ("in situ" is Latin for "in place"). Using drilling technology, steam is injected into the deposit to heat the oil sand lowering the viscosity of the bitumen. The hot bitumen migrates towards producing wells, bringing it to the surface, while the sand is left in place.

The most promising thermal recovery technology for Canadian bitumen resources is the steam-assisted gravity drainage (SAGD) process. In this process, two horizontal wells aligned in a vertical plane are drilled near the bottom of the reservoir. The top horizontal well is used to inject steam, while the bottom well is used to collect the produced liquids (oil, condensed steam, formation water). The steam injected from the top well rises into the formation, forming a large steam chamber above the well. The rising steam condenses on the boundary of the chamber, heating the oil and allowing it to drain to the production well. The process leads to high oil rates and a high recovery of the original oil in place at economic oil-to-steam ratios.

The SAGD process is energy intensive. For some reservoirs, the vapour extraction (VAPEX)rocess may have more potential. It is an analogue to SAGD, involving the injection of vaporized solvents (e.g. ethane, propane, butane) or solvent/gas mixtures rather than steam. This process results in the formation of a vapor chamber rather than a steam chamber, but with a similar reduction in oil viscosity at the chamber boundary that allows the oil to drain to the production well. Although VAPEX is less energy intensive than SAGD, production rates are also lower.

In addition, hybrid steam/solvent processes are currently under development for reservoirs in which steam or solvent processes alone are not suitable.

Production from in situ already rivals open pit mining and in the future may well replace mining as the main source of bitumen production from the oil sands. Challenges facing in situ process are efficient recoveries, management of water used to make steam, and co-generation of all (otherwise waste) heat sources to minimize energy costs.

Figure: Oil sands

5.2.4 Natural Gas

5.2.5 Coal

5.3 Mining

The CIMS mining model represents metal mines as well as potash mining. Other non-metal mines are not modelled in CIMS, as they currently consume a relatively small percentage of total mining energy. Activities related to coal mining and oil and gas extraction are covered elsewhere in CIMS.

Metal mines are typically categorized as either open-pit or underground. While the general production processes that occur in both categories are about the same, specific aspects of the mining technologies can differ significantly. For example, underground mining operations must address air quality issues in the mine shaft, which may require: cooling, heating and ventilation.

The basic processes for metal mining include most of the following generic steps (although not always in this order): initial breaking, transport (hoisting, trucking, conveying), crushing, grinding, concentration (mineral separation) and waste disposal (rock and tailings). Energy consumption in the final smelting and refining stage is not included in this sub-model but has been included in the CIMS metal smelting model.

5.4 Electricity

TODO: Competition type to aggregate demand load curves, sort, and assign to base, shoulder, and peak based on rules TODO: Peak shaving policy via load curve multiplier (<1 for peak, >1 for off-peak)?

The electricity model in CIMS is structured to account for utility-generated electricity. Electricity co-generated by industry is accounted for separately in the industrial sub-models. Please note that all electricity generated from renewables is accounted by the electricity model no matter whether it is generated by a utility or an industrial establishment.

Final electricity demanded in the entire Canadian economy (excluding co-generated electricity by the industry) is represented by the primary node (Electricity). The amounts of electricity allocated to intermediate and competition nodes flows directly from this primary node, which is the principle driver of the simulation runs. Intermediate nodes in the Electricity sub-model reflect the specific characteristics of electricity demand. Electricity demand varies significantly reflecting the fluctuating demand requirements depending on time-of-day (i.e. daily demand typically peaks between 4-6pm) and by time-of-year (i.e. annual peak demand during the cold winter months). Utilities track these varying demand conditions using load curves in an effort to supply electricity on a continual basis.

Electricity generation technologies have traditionally been classified based on their relative variable costs and the resulting frequency with which they are called upon to meet electricity demand or ‘’load‘‘; e.g.,‘’baseload,’’ ‘’load-following,’’ and ‘’peaking’’ resources. This classification is no longer meaningful in power systems with substantial penetration of wind and solar energy, since dispatch of each technology is also driven by the irregular variability of these renewable resources. Moreover, most available low-carbon technologies are capital intensive and have very low variable costs. In this context, the distinguishing attributes of electricity technologies relate more to their resource availability and ability to adapt production output in order to meet instantaneous demand.

1. ‘’Fuel-saving’’ variable renewable energy (VRE) resources. These include wind power, solar photovoltaics (PV), concentrating solar power, and run-of-river hydropower. They harness renewable energy inputs (wind, solar insolation, water) that vary on timescales ranging from seconds to hours to seasons, have zero (or near-zero) variable costs, and have no fuel costs. At lower penetration levels, they may displace the need for firm capacity, but, at higher shares, capacity needs are driven by periods with low VRE availability. At high energy shares, these technologies therefore contribute value primarily by displacing other higher variable cost generating technologies whenever available and reducing the total fuel consumption and variable costs of the system.
2. ‘’Fast-burst’’ balancing resources. These include short-duration energy storage (e.g., Li-ion batteries), flexible demand (or schedulable loads), and demand response (or price-responsive demand curtailment). They are either energy constrained (storage, demand flexibility) or have very high variable cost (demand curtailment). These technologies are

therefore poorly suited to operating continuously over long periods of time and are better used during high-value periods when relatively fast bursts of power or quick demand adjustments are needed to balance supply and demand.

3. ''Firm'' low-carbon resources. These are technologies that can be counted on to meet demand when needed in all seasons and over long durations (e.g., weeks or longer) and include nuclear power plants capable of flexible operations, 12–15 hydro plants with high-capacity reservoirs, coal and natural gas plants with CCS and capable of flexible operations, geothermal power, and biomass- and biogas-fueled power plants.

Aggregating and rearranging the daily load curves in descending order from highest demand (peak) to lowest demand yields a load-duration curve (LDC). An LDC relates the capacity required with the capacity utilized, in other words the number of operating hours required during the year for each proportion of peak capacity demand. The LDC is divided into time segments (i.e. hourly or daily) with the extreme left side representing the province's peak demand for the year (i.e. usually a winter evening) while the extreme right side represents the baseline power demand (i.e. the minimum amount of power required). Interpretation of the LDC allows for the determination of capacity i.e. represented by the height of each slice, and of the utilization rate i.e. the width of each slice. The product of these two provides'a measure (estimated) of electrical energy (e.g. kilowatt-hours).

As illustrated by the energy flow model of the electricity industry, total electricity generation in CIMS is split between the base, shoulder and peak nodes using ratios based on historical load curves. These ratios can change over time but are always user-specified.

Baseload is a utility's minimum load over a given period. A baseload generating facility provides the basic amount of electric power needed year-round except for maintenance and scheduled or unscheduled outages; normally a large, efficient power plant having a low cost per kilowatt-hour generated. The base node includes conventional technologies such as gas turbines, combined cycle gas turbines, simple cycle coal turbines, pulverized fluidized bed coal, integrated gasification combined cycle coal, small and large hydro, nuclear; and renewable technologies such as biomass steam plant, fuel cell generator, parabolic trough solar generation, geothermal, solar thermal and wind generation.

The shoulder and peak nodes only contain conventional technologies. Moreover, as a peak node plant needs to be phased in and out of the grid relatively quickly, plants with long start time such as nuclear plants could not be used as peak node facilities

While the split among base, shoulder and peak nodes are always user-specified (decision rule: fixed_matrix_ratio), the split between conventional and renewable generation in the base node can be either user-specified over time or based on the aggregate relative annual levelized costs of the technologies on each sub-node.

The current CIMS model deals with electricity imports and exports through the macroeconomic module. The own price elasticity for electric power export demand is -1.07. And the energy trade elasticities are applied using the following equation:

(Equation 31)

Where:

FLCCt = Current financial lifecycle cost for a product

FLCCt-1 = Previous financial lifecycle cost for a product

= % of market price covered by sub-model

= Own price demand elasticity for the energy commodity

= Corrected demand multiplier for the given market

Figure: Energy Flow Model of Electricity Generation

Rates

5.5 Ethanol & Biodiesel

5.6 Hydrogen

5.7 Industrial Sector

TODO: Break out temperature demands for key industries

low temp -> heat pump fuel switching, med (?), high -> need combustible fuel?

Check Chris Bataille's database for this.

This document describes how the industrial sector is modelled in CIMS. We provide a general description of the sector, its energy end-use processes, and the salient features of the model. The prototype for this CIMS module was the ISTUM-I model developed at Simon Fraser University.

5.7.1 Background

Unlike the commercial and residential sectors, the industrial sector is too diverse to simulate in a single module. CIMS therefore includes the following industrial sector sub-models, each reflecting the activity of one major industrial branch: chemical manufacturing, industrial minerals, iron and steel, mining, metal smelting and refining, petroleum refining, pulp and paper, and other manufacturing. Coal mining and oil and gas extraction are covered in the CIMS energy supply component. In some cases, a branch may include more than one industry (e.g., chemicals includes many unrelated chemical-producing industries, non-metallic minerals includes cement, and in some regions, lime, glass and brick production) while for some branches, the model is industry specific (e.g., petroleum refining, pulp and paper).

In 2003, the industrial sector consumed 2,733.5 PJ of secondary energy, or 32% of the energy used in Canada. Figure 7 provides the breakdown of total industrial energy use to the sub-branches represented in CIMS. Pulp and paper consumes more than any other industrial branch, at 30% of the total. Other manufacturing and petroleum refining follow, at 20% and 14% respectively. Metal smelting and refining and iron and steel each account for about 10% of industrial energy consumption. The remaining 10% is attributed to mining, industrial minerals, forestry and construction (the last two categories are referred to as “other” in the figure).

Figure 7. Energy Use by Industrial Branch in Canada in 2003

Source: Natural Resources Canada. 2005. Energy use data handbook, 1990 and 1997 to 2003. Ottawa: Office of Energy Efficiency, Natural Resources Canada.

Some industries are found in very few regions (e.g., aluminum smelting occurs only in BC and Québec) while others occur in most regions (pulp and paper occurs in all regions except Saskatchewan). Table 12 shows the disaggregation of industry sub-sectors in CIMS by region.

Table 12. Disaggregation of Industry Sub-sectors in CIMS

Each industrial sub-model in CIMS has its own driving variable, usually the total amount of final product produced or the amount of raw input processed (e.g., tonnes of steel, tonnes of mineral ore throughput, m3 refined petroleum products). The driving variables are determined exogenously, often based on official forecasts used by Natural Resources Canada to develop Canada’s Emissions Outlook: An Update, although in several cases sector specific sources are used.

In CIMS, the product and energy service demands in a sub-sector are linked in a flow model that describes the sequence of activities required to generate that product. The energy flow model for the iron and steel industrial branch is shown in Figure 8 below as an example. A CIMS flow model is geared towards representing technology evolution and energy consumption rather than economic criteria (as in an econometric model where units are typically in monetary terms) or actual mechanical processes (as in the blueprints or process flow diagrams used by engineers). Because the emphasis is on energy consumption and not material flow, the nodes in the flow model represent process stages in which energy consumption can be distinctly estimated. The flow model describes hierarchical nodes, linked by engineering ratios. Technology competitions take place at the lowest level nodes in the hierarchy.

Figure 8. Energy Flow Model of the Steel Products Industry

Nodes shaded in green represent unique industry processes, while nodes shaded in other colors represent auxiliary systems. These are generic energy services that are supplied to the major process technologies and are shared by all CIMS industry sub-models. The auxiliary systems fall into four general categories: steam generation (boilers and cogenerators), lighting, heating, ventilation and air conditioning (HVAC), and electric motor systems including pumps, fans, compressors and conveyors. Figure 9 shows the energy flow diagrams for auxiliary systems not included in Figure 8. In some cases, the energy service meets the direct need for steam, pumping or compression, while in other cases, it serves only to provide suitable conditions for production to continue, as in the case of lighting and HVAC systems.

Figure 9. Auxiliary Flow Model Diagram

The sections that follow describe how key auxiliary systems and process specific components are captured by the CIMS industrial model. The section on auxiliary systems covers steam generation and electric auxiliary services. Lighting and HVAC are not described in detail because the energy demanded by these services is small. The section on process specific systems covers each of the CIMS industrial sub-models.

5.7.2 Auxilliary Systems

5.7.2.1 Steam Generation

A number of industry process technologies generate steam as a by-product. CIMS accounts for steam generated by these technologies in the overall steam demand in each industry. Any shortage of steam must be met by boilers and cogenerators.

Boiler efficiency can vary greatly depending on boiler design, age, and fuel used. For modern oil and gas boilers, thermal (first law) efficiencies may be 85% or higher. Boiler efficiencies can be increased by introducing non-condensing and condensing heat recovery systems and by installing regenerative burners with computerized fuel/air mixtures to maximize fuel efficiency.

Cogeneration is the sequential production of electricity and useful thermal energy, usually as steam or shaft drive power. In cogeneration a boiler is used in conjunction with a turbine system to generate electricity, with the “waste” steam going to meet process steam requirements. In comparison to conventional electricity generation, cogeneration facilities are capable of conversion efficiencies of up to 85% (including both the steam and electricity produced) as compared to utility condensing turbine systems which have an efficiency in the 30% to 40% range.

5.7.2.2 Electric Auxiliary Systems

The vast majority of electricity consumed by industry is used by motor systems. A motor is the core component of a much broader system of electrical and mechanical equipment that provides services, including hydraulic power, compressed air, motive power and air flow. Opportunities for efficiency improvement exist in both the motor itself, and in the latter systems – pumping, air displacement, compression, conveyance as well as other types of machine drive that are unique to a given production process (direct drive).

Figure 3.3 above is a generic energy flow model representing motor and steam auxiliary systems as they appear in the various CIMS industrial sub-models. As the figure indicates, CIMS simulates six auxiliary groups related to process drive (in red). The first five demand shaft or mechanical drive provided by the sixth group, motors (machine drive). Each group simulates technology evolution of a specific system.

Increased disaggregation in some of the groups reflects differing end-uses and levels of efficiency. For example, compressor systems vary in electrical efficiency by size. Therefore, competition between different technologies must be separated by size. The energy flow model reflects this by displaying two sizes of compressors, which are available for use by the model.

Pumping, air displacement, compressing and conveyance systems have been represented as packages within CIMS. The components of each of these systems are associated with the key end-use. For example, a pipe, throttle and speed control device is attached to a pump, and the entire package is called a pump system. In the model, there are at least two types of systems:

1. The least efficient system comprised of the least efficient components
2. The most efficient system comprised of the most efficient components

The system efficiency is then a function of the efficiency of all the components within that system. In some cases intermediate efficiencies have also been listed. New technologies may be added to reflect the intermediate range.

Electronic variable speed drives (VSDs) provide speed control by varying the rotating speed of the induction motor to match variable speed requirements of driven equipment. Pumps and fans, for example, are often required to deliver varying qualities of fluid flow. When this control procedure is used the energy consumption of the fan or pump is dramatically reduced during part load operation, as compared to the conventional practice of using a valve or damper to reduce flow.

5.7.3 Petroleum Refining

This industrial sector includes establishments primarily engaged in manufacturing petroleum products. Activities such as the manufacture of lubricating oil and grease or asphalt and coal/coke products are included in CIMS where they are carried out in these plants. Establishments that are primarily engaged in these activities are not included; however, because of their comparatively small size.

The basic process involves refining crude oil into a number of products including: ethane, propane, butane, motor gasoline (typically 40% of total product), naphthas, jet fuel, petrochemical feedstocks, distillate fuel oil, diesel fuel, residual fuel oils, lubricants, coke and asphalt. All crude includes these products in varying concentrations. However, it is possible to increase output of more desirable products per unit crude input by using cracking and reforming processes.

The type and quality of crude and the processing requirements to generate the end products determine the refinery's complexity and have significant impact on energy consumption in the plant. In spite of this, however, the basic processing procedures are common to all refineries. The major process steps include: atmospheric distillation, vacuum distillation, cracking, desulphurization, alkylation, isomerization, reforming, sulphur recovery and hydrogen production. The last two processes should not be considered part of the actual refining process; they act as utility functions to clean out sulphurous compounds or provide hydrogen for specific in-plant operations.

Figure: Energy Flow Model for the Chemical Products Industry

5.7.4 Industrial Minerals

The industrial minerals sub-model of CIMS generally includes all industry involved in the production of all non-metallic mineral products. It encompasses the cement, lime, glass and brick industries, with cement and lime as the major energy consumers. These key products both require significant quantities of thermal energy per unit production during the calcining process. Other energy-demanding processes involved in cement or lime production typically include raw materials preparation (this excludes extraction of the raw materials from their in situ locations, even though these may be within the plant gate), and finishing or finished materials preparation. Calcining remains the most energy intensive part of the processes, consuming large quantities of combustible fuel (85% or more of the total energy consumed). Initial and final preparation of the materials requires electricity to drive motors that are attached to various auxiliary and grinding devices.

Figure: Energy Flow Model for the Chemical Products Industry

5.7.5 Iron and Steel

The CIMS iron and steel model represents this industrial branch as a process whole, including the production of coke from coal, the incorporation of iron ore (including any agglomeration that occurs on site) and ending with the set of end products. End products include the three basic forms (slabs, blooms and billets) and, in many cases, their sub-forms. The iron and steel industrial branch has a high energy intensity per tonne of product.

Presently, two different processes generate Canadian steel. In the first process, coke, a coal derivative, reduces iron oxides in ore to pig iron in a blast furnace. Basic oxygen furnaces (BOFs) then purify this liquid iron along with some scrap by injecting high purity oxygen, which is itself an energy-intense product. In the second process, electric arc furnaces (EAFs) recycle 100% scrap metal. In both cases, the molten steel (with carbon content less than 2%) may be cast, inspected, reheated and finished. Each of a wide variety of finished products requires varying inputs of heat and mechanical energy to its manufacture.

The steel industry is rapidly modernizing. Most of these technologies concentrate on energy-saving measures and reuse of what had historically been waste heat. Direct reduced iron and direct smelted iron processes, presently in pilot scale plants, may eventually make the coke-dominated process redundant. More advanced finishing processes such as direct rolling and thin slab and thin strip casting will eliminate reheating of steel (typically done after initial inspection).

Figure: Energy Flow Model for the Chemical Products Industry

5.7.6 Metal Smelting and Refining

The metal smelting and refining model of CIMS includes establishments that are primarily engaged in manufacturing finished metal products excluding iron and steel. These processes consume very high amounts of energy per unit of product. The CIMS sub-model explicitly includes processes related to aluminium, nickel, copper, zinc, lead, magnesium and titanium. Though economically important, minor elements such as gold, silver, platinum and cadmium are not represented explicitly because they are often processed in conjunction with the metals listed or are processed in too small a quantity to require direct representation.

In metal smelting, metal is typically extracted from the concentrate by leaching (hydrometallurgical recovery) or through heat (pyrometallurgical recovery). In Ontario, for example, all zinc smelters use hydrometallurgical techniques while copper and nickel smelters use pyrometallurgy. Smelting processes generate products that contain between 95% and 99% pure metal. Refining, if it occurs, depends on the type of smelting technique used. Products prepared through hydrometallurgy tend to use electrowinning for refining while products prepared through pyrometallurgy tend to use electrolysis or fire refining. Refined products are at least 99.99% pure metal.

Purification of the aluminum is accomplished utilizing the Hall-Heroult process, which requires large quantities of electrical energy. Although Canada has no aluminum ore body of sufficient concentration to warrant extraction, readily available, inexpensive electrical energy attracts companies who wish to refine the metal from alumina (concentrated ore). Recycled aluminium requires only 5% of the energy of original production. Increasing demand for conservation and recycling may have a significant effect on the production of virgin metal.

Figure: Energy Flow Model for the Chemical Products Industry

5.7.7 Chemical Products

Unlike some of the other industrial sectors modelled in CIMS, the chemical products sector is not dominated by a single energy intensive product to which all others can be associated. As a result, several products and processes must be modeled to accurately reflect the energy flow within the industry. The dominant chemical products, from an energy consumption perspective, are: chlorine, caustic soda (sodium hydroxide), sodium chlorate, hydrogen peroxide, ammonia, methanol, ethylene, propylene and polymers. Sub-nodes in the model represent different process stages in which energy consumption can be distinctly represented.

Ethylene, methanol, toluene and xylene are the primary petrochemicals produced. Crude oil and natural gas feedstocks provide the raw material from which these are produced. Ammonia, chlorine (often produced in complement with caustic soda) and sulphuric acid are the major inorganic chemicals produced. Like petrochemicals, natural gas provides the principal raw material used in the manufacture of ammonia. Ammonia, in turn, forms the primary feedstock for urea, ammonium phosphate, ammonium nitrate, ammonium sulfate and nitrogen solutions commonly sold as fertilizers.

Figure: Energy Flow Model for the Chemical Products Industry

*This flow model has not yet been implemented in CIMS

5.7.8 Pulp and Paper

TODO: Check with Michael at Navius about JCIMS data and model for CFS project in 2017.

TODO: Need to early retire biomass steam generation stock to be able to meet emissions in 2010-2020

The pulp and paper industry includes establishments primarily engaged in manufacturing pulp, paper, paperboard and building and insulation board. These establishments are included in the CIMS pulp and paper sub-model because they are large in size and their products are highly energy intensive when compared to other industry products. Paper products, which do not form part of the sub-model include: asphalt roofing, paper box and bag, and other converted paper product industries. These are included in the other manufacturing model of CIMS. The pulp and paper industrial branch has a high energy-intensity per tonne of product.

Pulping processes can be broken down into three basic types: chemical, mechanical, and recycled. Mechanical pulping, which tends to be very electricity intensive, uses wood fibre much more efficiently than does chemical pulping. Recycled pulp consumes considerably less energy than either of the other two major processes but input supply is limited and the collection and transportation of this supply can significantly increase the energy intensity of the product. The products generated from each of these processes are distinct enough that they generally do not compete on the open market.

Each pulp product serves as the stock for many types of paper products. Newsprint, the largest category in terms of production (> 60%), can use feedstock from each of the pulping processes. Other paper products incorporated into this sub-model includes tissue paper, uncoated and coated paper (or wood-free paper), and linerboard, each of which may have varying degrees of stock inputs.

The following steps in the production of pulp and paper are common to both chemical and mechanical mills where raw logs serve as feedstock: wood debarking, wood chipping, pulp making, pulp washing, pulp bleaching, pulp drying, paper stock preparation, paper sheet formation, paper pressing and paper drying. Mills that use chips or recycled paper as feedstock may exclude some of these steps.

Figure: Energy Flow Model for the Chemical Products Industry

5.7.9 Other Manufacturing

The CIMS industrial sub-model for other manufacturing industries has been created to capture those industries within a region, which, on their own, do not consume enough energy to merit the development of a separate model. When considered as a group, however, they may consume a significant (if not the dominant) portion of the energy within any one region. The industries considered part of the other manufacturing sector are: food, beverage, tobacco, rubber products, plastic products, leather and allied, primary textiles, textile products, clothing, wood, furniture and fixture, printing and publishing, fabricated metal products, machinery, transportation equipment, electrical and electronic, and other manufacturing.

The other manufacturing group of industries typically includes a large variety of technologies and processes. Usually, a simplified flow model of generic energy services represents adequately the energy consumption of these industries as a whole. Steam boilers, process heat, space heat, and electricity for lights and electric motors and their attached auxiliary devices (pumps, conveyors, fans and the like) constitute the bulk of energy-using services.

Unlike other "major" industries the other manufacturing industry as a group does not have any single product which dominates the production processes and to which the production of all other products may be linked. As a proxy for the physical output measures used in the sub-models major industries (i.e., tonnes of steel or cubic metres of gasoline) the output of the other manufacturing industry is represented in monetary terms such as the RDP (real domestic product) of the industry.

Figure: Energy Flow Model for the Chemical Products Industry

5.8 Direct Air Capture

- Run year as normal with carbon price
- Re-run year with shadow carbon price, iterate until between:
  - total emissions difference between runs is equal to maximum DAC quantity
    - have maximum annual growth rate for DAC?
    - Expert opinion?
  - minimum shadow price equals DAC cost
- Emissions difference between runs is DAC request by tech
  - Only change shadow price in sectors that are allowed to purchase DAC
- Total emissions difference is DAC sequestered

5.9 Residential

TODO: Track hot water usage through sector using L of hot water (currently m3 I think), then change to efficiency at hot water node (although, check that we can find heating efficiency for hot water production).

TODO: Check best units to convert from clothes washing to drying. Should be something like loads/cycles per year (i.e., more efficient washer reduces need for drying, but can be combined with more efficient drying).

This section describes the structure and data inputs of the residential module of CIMS. It provides a general description of the sector its energy end-use processes, and the salient features of the model. The model was initially developed as ISTUM-R by Alison Bailie (1994).

5.9.1 Background

The CIMS residential sector sub-model attempts to capture all energy demand by residential buildings within Canada. This includes single family dwellings, duplexes, row housing, walk-ups and large apartments. Large apartment buildings are often considered commercial buildings by utilities, but, since their energy consumption patterns better fit within the residential sector, they have been included as part of this sub-model.

The number of households drives the residential sector models and is calculated exogenously, based on future population growth estimates. This variable directly determines the demands for all services, except for water heating and furnace fan demand. The technologies for services that require hot water and furnace fans establish those demands.

5.9.2 Energy Use in the Residential Sector

In 2002, the residential sector accounted for approximately 17% of all energy consumed in Canada. This energy provides services such as space heating, water heating, lighting, cooking, refrigeration and clothes drying. Space heating accounts for about 60% of residential energy consumption, water heating approximately 21% with the remainder going to lighting, appliances and other uses. The exact nature (i.e., quantity, fuel type etc.) of this consumption depends on many variables such as availability, climate, building type and occupancy patterns.

The following describes energy end-use services common to residence types incorporated into the model, the variables that affect the amount of energy consumed in providing the services, the relationship between the end-use and the driving variable at the primary node (number of households) and any other interrelationships that permit a more accurate simulation of the end-uses.

5.9.2.1 Space Heating

A variety of factors uniquely determine the energy consumption required to heat a household. These factors include the physical building envelope, the type of heating system, maintenance of both the envelope and heating system, climate and exterior landscape features, and thermostat settings. Modelling this service requires many assumptions and generalizations. Due to the multitude of housing types, the sector must be condensed into groups consisting of thousands of households which are represented in the model by one typical house archetype. All heating systems cannot be included, due to the large number of possibilities; thus the models contain only the most common systems and those considered likely to gain significant market shares in the future. Case studies and surveys provided estimates for geographical and behavioural assumptions.

Each space heating technology in the residential CIMS sub-model combines a shell archetype and a heating system. Each archetype, which is a set of physical measurements for components such as floor space, window area, insulation levels, and rate of air exchange, represents a group of households. Modellers use archetype descriptions to calculate the heatload and cost of the typical building as well as the cost and energy savings of retrofitting. For each shell archetype, CIMS includes a choice of four or five different heating systems. These systems include oil furnaces, natural gas furnaces (typically three efficiency levels), high efficiency space heat/hot water systems, wood stoves, electric baseboard heating, and electric air source heat pumps.

5.9.2.2 Non-Appliance Hot Water Use

This service meets the hot water required for baths, showers, hand washing, and cleaning. The competing technologies, shower heads and faucets with varying flow (litres/minute) levels, consume hot water rather than energy. The water demand determined by this node contributes to the hot water demand that drives the hot water heating service.

5.9.2.3 Lighting and Major Appliances

Each of the following appliances is modelled as a separate service:

- Refrigerators
- Freezers
- Dryers
- Ranges
- Dish Washers
- Clothes Washers
- Air Conditioners
- Lighting

The stock is measured in number of appliance units (i.e., number of refrigerators) and the ratio between the service and the driving variable (number of households) is the market penetration of the appliance. Generally, there is a range of efficiency levels for new stock in each major appliance service.

Clothes washers demand both electricity and hot water. Dish washers are disaggregated into two categories: machine, which demand electricity and hot water, and non-machine, which only consume hot water. Due to varying demand for air conditioning not all the residential sub-models incorporate them as a separate end-use and they are often combined with the minor appliances.

5.9.2.4 Minor Appliances

This service contains many diverse technologies including televisions, car block heaters, lawn mowers, and computers. Although, these appliances collectively consume a significant portion of household energy, individually either their consumption or their market penetration is low. As a group, this end-use is growing considerably. Efficiency improvements are outweighed by increase in number of goods.

5.9.2.5 Renewable Energy Sources and Other Distributed Energy

A number of renewable technologies (in addition to wood stoves) can be applied at the household level to provide both passive and active energy. These include photovoltaic panels, geothermal, solar hot water heating and passive solar design. Options are also increasingly becoming available to generate electricity using combined heat and power generations at the household level, for instance through fuel cells and through micro-CHP (for instance the stirling engine). Some of these options are modelled at the end-use in which they meet thermal demand, rather than as a unique 'supply' node in the flow model.

5.9.2.6 Water Heating

The water heating demand depends on the technologies used to supply services of non-appliance hot water, dish washing, and clothes washing. For most purposes, hot water must be at 60oC and, unheated water generally enters the household at 15oC. Both storage tank and non-storage tank water heaters are modelled. Fuel cell and stirling engines are modelled at this end-use to meet water heat loads (while generating electricity for household use).

5.9.2.7 Energy Flow Model, Residential Sector

Although each province in Canada has a number of unique characteristics in terms of climate, the availability of fuel and the structure of the residential sector, all households require the same basic set of energy services. The following section describes the basic structure and the end-uses common to all regional residential models. Major differences between regional models are described later in this section.

Figure 5 presents CIMS's residential sector flow sub-model. The number of residences required in any year drives the model. The flow model demonstrates the linkages between the driving variable and the energy end-uses in the residential sector. The number of residences is determined exogenously, based on future growth estimates of population and economic activity. The end-uses are linked to the driving variable through engineering ratios based on historic relationships between a residence and each end-use.

5.9.2.8 Energy End-Uses (Non-Space Heating)

Due to similarities in service demand and technology choices, the lighting and major appliance services modelled in the residential sector have been aggregated for all housing types. Air conditioning, a large energy consumer in some regions (e.g., Ontario), consists of room or window units and central units. Disaggregating apartment and non-apartment water heating captures the level of use and technology differences between these two categories.

5.9.2.9 Space Heating

In contrast to the other energy services in the residential sector, the amount of space heating demanded by a residential unit can vary considerably depending on the building type. As a result, the space heating service node of the model is divided into three basic housing types: apartment, attached (row), and single family detached dwellings. The apartment category includes both high and low rise apartments while the attached category consists of duplexes, triplexes, and townhouses. As described above each of these housing types have been divided up into different archetypes to describe the differences in building heatloads and the fuels and efficiencies of the heating systems.

5.9.2.10 Apartments

In the apartment node, two basic technology archetypes are available, standard and 'improved shell'. The standard archetype generally represents the existing or a baseline shell design that demands a relatively high heat load while the improved apartment shell design represents better insulation, windows, etc., demanding a lower heat load. Each of these archetypes is divided into several different heating system types based on the type of fuel and efficiency of the heating system, e.g., oil heating, several efficiency levels of natural gas heat (62%, 78%, 90%) and electric heat. During each model run new apartments can be purchased either from the options available from existing stock or from a selection of new buildings with an improved shell design and better heating systems. There are no retrofit options currently available for existing apartment stock.

5.9.2.11 Attached Housing (Row)

The attached housing category is modelled in the same manner as the apartments.

5.9.2.12 Single Family Dwellings

The greatest level of technological variety in the residential sector exists in the types of systems for heating single family dwellings (SFDs). These technologies also show the greatest differences between the regions. In the basic CIMS residential sub model, SFDs are divided up into two categories, existing and new housing.

Two archetypes are used to model all existing houses. Archetype A houses have very little insulation and, therefore, higher heatloads than Archetype B houses. Where wood consumption is present, wood stoves meet part of heat load demand. CIMS allows houses in either of these archetypes to retrofit to improve insulation levels. The six possible heating systems for the current housing stock are oil furnace, electric baseboard heating, air-source heatpump, standard natural gas furnace (78% efficient), high-efficiency natural gas furnace (78% efficient), and high efficiency space heat/hot water systems.

In each region, three types of houses compete for new stock: standard, improved and R-2000 archetypes. R-2000 houses meet the current heating budget specified by R-2000 standards for the region. The improved archetype is based on the ratio between standard and improved archetypes. Houses can be upgraded to better insulated categories through retrofits. Heating systems are the same as those modelled for existing homes. In some regions wood stoves may be chosen to heat standard and improved archetypes.

Figure 5. Energy Flow Model for the Residential Sector

5.9.3 Regional Differences in CIMS Residential Sub-models

5.9.3.1 British Columbia

The BC residential sub-model has some differences compared to the other provinces in Canada. t is the only model to make use of the primary/secondary climate distinction available for single family detached homes - coastal SFD (Lower Mainland and Vancouver Island) and interior SFD are represented. The same has not been done for apartment and attached housing since over 80% of these buildings are in the coastal area and this trend is assumed to continue over the modelling period.

The coastal SFD (Primary Climate) housing portion of the model is as described above. The interior region (Secondary Climate), with only half as many houses as the coastal region, uses only a single archetype to represent existing stock. The archetype heatload is the average of all current housing stock heatloads. These houses, like the others, can be retrofitted to improve insulation levels. Since wood provides primary heat for approximately 20% of interior houses, CIMS includes wood stove heating systems for these archetypes. The other heating systems in the interior are identical to the coastal systems described above.

5.9.4 Climate Zones

5.10 Commercial sector

TODO: Split auxiliary end uses by activity types (e.g., lighting, hot water, etc.)?

5.10.1 Background

5.10.2 Energy Use in the Commercial Sector

In CIMS, the definition of the commercial sector excludes light industrial facilities, which are included in the other manufacturing sub-model, and high-rise residential apartments, which are included in the residential sub-model.

The commercial building sector accounts for roughly 20% of all energy consumed in Canada. For the most part, energy in the commercial sector is consumed to provide energy services such as lighting, heating, ventilation and air-conditioning (HVAC); cooking, refrigeration, hot water, and plug load (i.e. power for computers and other office equipment). Lighting and HVAC account for roughly 60% of the energy consumed in this sector. However, the distribution of energy use in any particular commercial building depends on its architectural characteristics, building type and size, efficiency of installed equipment and activities of the occupants.

The most common fuels used in the commercial building sector are, in order of importance, electricity, natural gas and light fuel oil. Electricity is the only energy source used for lighting, refrigeration, plug load, cooling, fan and pumping. Electricity and natural gas are used for cooking. All three energy sources are used for space and hot water heating.

5.10.2.1 Energy Flow Model, Commercial Sector

Figure 6 shows CIMS's commercial sector energy flow model. CIMS considers commercial floor space as being the driving variable (primary node) of the commercial sector. The flow model demonstrates the linkages between the driving variable (commercial floor space) and the energy end-uses in the commercial building sector. The amount of commercial floor space is determined exogenously, based on future growth estimates of population and economic activity. The end-uses are linked to the driving variable through engineering ratios based on historic relationships between commercial floor space and each end-use. Not all energy end-uses found in the commercial sector are included in the model. Only space heating, lighting, refrigeration, cooking, hot water and plug load are considered significant enough, in terms of their total energy consumption, for inclusion.

End-uses are described in more detail in the next sections.

Figure 6. Energy Flow Model for the Commercial Sector

5.10.2.2 Lighting

In the commercial building sector, lighting consumes more electricity than any other end-use, accounting for 30 to 40 percent of total consumption. Lighting is measured in terms of the number of square meters of commercial floor space that it serves. Therefore, the lighting service has a one to one ratio with commercial floor space.

Lighting is not modelled separately by building segment. CIMS simulates lighting at different energy efficiency levels. The base technology represents the average efficiency of stock in 2000. The 'new' technology represents the marginal energy efficiency of new stock built in 2000. Other options represent actions to adopt more efficient lighting equipment (super T8, advanced HID, compact fluorescent lighting) and improving lighting design.

The lighting end-use category is modelled as three sub-categories: general area lighting, high bay lighting and service lighting. General area lighting is the dominant category of lighting and refers to the lighting found throughout general office areas and uses predominately fluorescent tube lights. Service area lighting refers to lighting systems used in corridors, lobbies, vestibules, washrooms, and any non-critical areas adjacent to areas illuminated by general area lighting. Lastly, service lighting refers to the lighting systems used in high bay areas of a building - large atria and lobbies, gymnasiums etc. The energy intensity of any sub-category depends on the efficiency of the lighting equipment and the use rate.

CIMS does not currently account for the energy consumed for exterior lighting such as street lighting, architectural lighting or parking lot lighting.

5.10.2.3 Shell/HVAC

The Shell/HVAC category is disaggregated into nine representative building segments (sub-categories). These segments, listed below, combine two or three different building types to group buildings with similar HVAC system types and usage. For example, the schools category includes elementary schools, colleges and universities.

- Warehouses and Wholesale Outlets
- Hotels and Motels
- Schools and Universities
- Small Office Buildings
- Large Office Buildings
- Small Retail
- Large Retail
- Hospitals and Nursing Homes
- Miscellaneous Buildings (not including residential or light industrial)

The mix of buildings in the segments is derived from data for the base year. The eight segments each have different occupancy and use rates, system types, and shell types. The relative shares of the building types within each segment can vary over the study period.

Each building segment encompasses a variety of shell and building HVAC technology options. Each is represented by a 'shell' node which competes alternative shell options. This represents a demand for heat load which is met by a corresponding HVAC service node for each building segment (see flow model). The link between the shell and HVAC competitions attempts to capture the relationship between decisions about building shell characteristics and the amount of energy demanded of the heating and air conditioning services. HVAC systems are modelled as 'single' technologies to represent the interdependence of choices about heating, cooling and ventilation systems.

5.10.2.4 Refrigeration, Cooking, Hot Water, Plug Load

The end-uses in the commercial sub-model are linked to the driving variable based on historic ratios of use per unit floor space and the weighted average of all the various commercial sub-sectors (e.g., a hospital will use more hot water than a warehouse per unit area). The weighted ratios can change over time, depending on changes in the proportion of the different kinds of floor space assumed over that period.

5.10.2.5 Plug Load

5.10.2.6 Domestic Hot Water

Domestic hot water (DHW) end-use represents the hot water used for all building services except space heating and is measured in cubic meters of hot water demand for the whole commercial building sector. It is related to the primary node by the ratio of cubic meters demand to commercial building floor space. Hot water is used for showering, washing, cleaning and food preparation.

5.10.2.7 Refrigeration

The refrigeration end-use includes all commercial refrigeration except for cooling required for air conditioning and is measured in gigajoules of refrigeration demand per square meter of commercial building floor space. This end-use is represented by two types of technologies; stand-alone plug-in refrigerators and large built-up refrigeration systems. These two types have different efficiency improvement opportunities. The refrigeration category in the model combines these two technologies and its energy intensity is the weighted average of the two system types. The ratio of the two system types is assumed to remain constant over time in a typical simulation.

5.10.2.8 Cooking

The cooking end-use is served by either electric or gas ovens, stoves and fryers and is measured in gigajoules of cooking demand per square meter of commercial floor space. This end-use also consists of different system types and the energy intensity of this end-use is the average for all types of commercial cooking equipment. As in the other end-use services, the shares of the various equipment types are assumed to remain constant over the simulation period.

5.10.3 Commercial Sector: Assumptions and Caveats

Any model is a simplification of reality. The results from CIMS are limited by data availability and the complexity of modelling all energy relationships in the commercial sector. The major assumptions and caveats made in the process of developing this CIMS sub-model are described below.

5.10.3.1 Energy Prices

As with the other CIMS sub-models, energy prices used in the commercial sector sub-model for both natural gas and electricity are based on the average charge per unit of energy consumed. Each unit of energy is assumed to cost the same regardless of the consumption rate of the customer. In reality both natural gas and electric utilities have rate structures that charge different prices depending on a customer's total consumption. Therefore energy cost savings predicted by CIMS may be different from actual savings.

Many utilities also have demand charges or fixed fees charged per kW of demand. Since many energy efficiency measures reduce the demand of a facility, they also reduce the demand charges. These cost savings have not been included in the savings predicted by CIMS.

5.10.3.2 Discount Rates

The main criterion used by CIMS for choosing technologies is the lifecycle cost of competing technologies. To reflect the constraints imposed on investments due to rapid payback requirements, risk aversion, capital availability and other investment criteria, annualized costs in the commercial sector are generally derived using a 40% discount rate. Cogeneration technologies, however, are discounted at a rate of 30% because investors in cogeneration are generally seen to have a longer planning / finance horizon than investors in energy efficiency. In addition, the hospital and school sectors' shell/HVAC technologies are discounted at a rate of 30%, instead of 40%, consistent with evidence of their longer planning / cost horizons and the fact that they are owner-operated. These rates reflect ex post measures of discount rates and are meant to encompass many more factors than simple rate-of-return criterion.

5.10.3.3 Retrofit Technologies

Some technologies are also set up as retrofits in the commercial sector. After existing stock is retired, CIMS compares the costs of the parent or lead technology and the attachable retrofit technologies. Although it remembers how much of the original parent stock there was, at this point CIMS assumes that there is no existing stock and sets up a competition between the lead technology and the retrofit technology(ies) to determine how much of that "remembered" original stock will remain as is (no retrofit) and how much will be retrofitted. The cost of the parent technology is simply the operating and maintenance (O&M) cost including fuel costs. The cost of each retrofit technology is O&M, fuel plus the annualized retrofit capital cost less any tax or other credit.

5.10.3.4 Interactive Effects Between Various Energy End-uses

The commercial building sector is different from both the industrial and the residential sectors in that there are significant interactive effects between energy end-uses and energy efficiency measures. As a result of these interactive effects, an energy efficiency measure that is undertaken for one end-use may have an impact on the energy consumption of another end-use. For instance, an energy-efficient lighting retrofit in a commercial building will reduce the cooling load but will increase the heating load because the waste heat from the lighting system is reduced. However, because there are so many factors that influence the consumption of energy by the HVAC system, these interactive effects are very difficult to quantify. Models exist that calculate these interactions for a specific building type but this type of analysis is beyond the scope of this current CIMS sub-model. The focus in the model is on capturing the main interactive effects between shell choice and HVAC demand.

5.10.3.5 Cogeneration

Cogeneration of heat and electricity has been limited exogenously in CIMS to between 30 and 50 percent of the demand in schools, large office, hospital, miscellaneous and hotel shell/HVAC categories. This is because cogeneration is only practical in large facilities that have a demand for both electricity and hot water or steam. The range of 30 to 50 percent is due to the different mix of buildings that make up each sub-category.

Cogeneration systems are sized to provide 100% of the facility's heating load. In certain instances this results in a surplus of electricity beyond that required by the shell/HVAC system. This surplus is assumed to be sold back to the electric utility for the same price that the facility is charged for electricity.

5.10.3.6 Oil Heating

CIMS assumes that no oil-fired heating is installed after 1990 in the commercial building sector of all regions except the Atlantic region, where natural gas is less available. This was done to simplify the model since oil heating is a small and declining portion of the market. The Atlantic region is the only region in Canada where the proportion of oil-fired heating is significant in the commercial building sector.

5.10.3.7 Lighting

CIMS does not account for the energy consumed for exterior lighting such as street lighting, architectural lighting or parking lot lighting. Data for these end-uses are not available.

5.11 Transportation

TODO: In model description

- Overhaul entire LDV sector
  - Setup ICE vs ZEV properly for all provinces – see BC or ON?
- Same for HDV
  - See Mariah's model
  - Move off-road out of freight to Residential (driver should be households (or population))
- Check fuel blending nodes
- Move load factors to DCC file
- Split air into domestic and international
- Add intercity ferry for BC trans personal

This document describes the structure and data inputs of CIMS-Transportation (CIMS-T), the transportation module of CIMS. It assumes that the reader is already familiar with the CIMS simulation algorithm and parameters. Additional documentation is available from the Energy and Materials Research Group that describes the general function of CIMS.

5.11.1 Service Flow Model

Figure 1 presents a simplified version of the service flow model of CIMS-T. The three main energy service categories are personal mobility, freight transport and off-road vehicles. In urban areas, single occupant vehicles (SOV), high occupant vehicles (HOV), public transit, walking and cycling provide personal mobility. Personal vehicles, buses, planes and trains provide transportation between urban areas. Marine, air, road and rail modes provide freight transportation services.

Figure 5 Service Flow Model of CIMS-Transportation

The SOV, HOV and personal vehicle nodes attached to the urban and inter-city nodes in the service flow model represent a demand for passenger vehicle services. The passenger vehicle service node located to the bottom left of the diagram supplies this demand. Personal vehicles are designated as either cars or trucks in CIMS. Within these categories, vehicles of different efficiency levels and fuel types (gasoline, diesel, natural gas, propane, methanol, ethanol, battery electric, gasoline-electric hybrid, hydrogen fuel cell) compete to meet demand. Efficiency and fuel type are specified only for new passenger vehicles. The nodes representing old and recent passenger vehicles are used simply to keep track of vintages of vehicles already present in the base year (2000). These vehicles are no longer available to supply new demand for passenger vehicle services. During the simulation, a retirement function removes these vehicles from the existing stock.

5.11.2 Endogenous Actions

CIMS-T has the capability to simulate the following actions endogenously:

- Changes in the efficiency of the gasoline personal vehicle fleet,
- Fuel switching within the personal vehicle market,
- Shifts between the personal vehicle and public transit modes,
- Shifts between SOV use and HOV use,
- Changes in the efficiency of the freight truck fleet, and
- Changes in demand for personal mobility.

With the exception of efficiency choices between road freight vehicles, technology shares within the freight sector continue to be simulated exogenously. The same is true for the off-road sector. Choices between public transit technologies (bus versus subway) are also specified outside the model. Actions that involve changes in the operation and maintenance characteristics of existing technology stocks are incorporated exogenously as the CIMS models simulate technological change through stock turnover, not through changes in the use and operating characteristics of existing stocks.

5.11.2.1 Input Data

The CIMS models are data-intensive: in order to carry out a simulation, input data are required to describe a series of characteristics for each technology. Technology characteristics and data sources for CIMS-T are listed in Table 1. The input data used to describe personal vehicles is described in more detail below, given the importance of this vehicle category to the transportation sector as a whole.

Table 1: Technology Specific Input Data for CIMS-T

Personal vehicles are defined by efficiency and type of fuel / technology. Technological characteristics of alternative fuelled vehicles were initially based on the output of the Transportation Issue Table. Vehicle cost differentials relative to conventional gasoline were taken from Levelton Engineering Ltd.'s work for the Issue Table (1999), with the exception of gasoline-electric hybrid and hydrogen fuel cell vehicles – for these vehicle types we used data from the NEMS model of the U.S. Department of Energy (1998). Energy consumption relative to conventional gasoline vehicles was also estimated based on the Levelton (1999) report. In 2004, a comprehensive literature review was undertaken to update the capital cost and energy consumption values for personal vehicles. Sources such as Azar, Lindgren and Andersson (2003); Ogden, Williams and Larson (2004); and Weiss et al. (2003 and 2000) were used to verify and adjust the existing data points.

The values of the intangible cost parameters within CIMS-T reflect differences between various efficiency levels and fuel types of SOVs, as well as between SOVs and other modes of transportation. The estimates of intangible costs for more fuel efficient vehicles are based on market studies of the revealed willingness-to-pay of many consumers for higher horsepower vehicles, which are generally less efficient (U.S. Department of Energy, 1998; AMG, 1999).

The estimates of intangible costs for alternative fuelled vehicles represent consumer preferences for longer range and wider fuel availability, as well as a general preference for conventional gasoline vehicles. The range and fuel availability parameters were derived by combining technology data of the U.S. Department of Energy (1998) with the results of a stated preference survey carried out by Bunch et al. (1993). The stated preference approach is necessary in dealing with alternative fuelled vehicles because these tend to be emerging technologies for which revealed preference data do not exist.

The estimates of intangible costs for public transit and HOVs relative to gasoline-driven SOVs are based on the financial costs and market shares of alternative modes of urban travel in Canada, as well as comparisons to various European countries where the financial costs associated with personal vehicle travel are much higher.

We are currently reviewing the intangible cost parameters used in CIMS-T based on data from a series of stated preference surveys carried out by the Energy and Materials Research Group and CIEEDAC.

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5.12 Waste

5.13 Agriculture

5.14 LULUCF

Need to include emissions for accounting. Want to have total emissions from CIMS equal the NIR for easy comparison.

Also include credits in GHG accounting, again for easy comparison. Or should this be done outside CIMS?

6 Decarbonizing Low-Temperature Industrial Heat

6.1 Scope and Definition

Low-temperature industrial process heat in CIMS is modeled as the steam, hot-water, and direct-heating service supplied to the industrial sub-models at end-use temperatures below approximately 200 °C. Process heat is partitioned into three temperature bands — low (below 200 °C), medium (200–400 °C), and high (above 400 °C). The bands are defined by end-use temperature rather than by fuel or equipment type, because the technology set admitted to the relevant service nodes differs substantially across them. The 200 °C boundary follows the convention adopted by the U.S. Department of Energy and applied in the recent peer-reviewed techno-economic literature (Schoeneberger et al. 2020). Industrial heat globally accounts for approximately two-thirds of final industrial energy use and roughly one-fifth of energy-related CO_2 emissions; analyses across IEA member countries place 30–45 % of industrial heat demand within the low-temperature band (International Energy Agency 2022, 2024; Madeddu et al. 2020).

The allocation of base-year low-temperature heat demand across sub-models on a provincial basis is the result of a calibration exercise bringing together information from Statistics Canada and Natural Resources Canada's Comprehensive Energy Use Database (CEUD), Environment and Climate Change Canada's National Inventory Report (NIR), and U.S. Energy Information Administration Manufacturing Energy Consumption Survey (MECS) data used for analogous subsector benchmarks. The CEUD does not publish process heat disaggregated by temperature band; end-use accounting nevertheless indicates that conventional steam systems, boilers, and process heating account for the majority of fossil-fuel consumption in Canadian manufacturing (Natural Resources Canada 2024).

6.2 Industrial Subsectors and End-Uses

Low-temperature heat demand is concentrated in a defined set of CIMS industrial sub-models — Pulp and Paper, Chemical Products, Mining, Industrial Minerals, Metal Smelting, Iron and Steel, and Light Industrial (which aggregates food and beverage, textiles, wood and prefabricated building products, and general light manufacturing). Each sub-model is calibrated independently for base-year energy consumption, and service demand is assumed to scale with industrial sectoral output projections in future years. The relative share of low-, medium-, and high-temperature heat within each subsector is assumed to remain constant in projection years. Subsectors and representative end-use temperature ranges relevant to the Canadian industrial mix are summarized in Table 6.1; note that food and beverage processing, textiles and laundry services, wood products, and light manufacturing and assembly are not standalone CIMS sub-models but are represented within the Light Industrial sub-model.

Table 6.1: Representative low-temperature end-uses and temperature ranges by industrial subsector

Subsector	Representative end-uses	Range (°C)
Food and beverage processing	Pasteurization, sterilization, spray and tray drying, clean-in-place wash water	60–200
Pulp and paper	Black-liquor evaporation, paper drying on steam-heated cylinders, digester heating	100–180
Chemicals and pharmaceuticals	Distillation reboilers for low-boiling solvents, batch reactor heating, crystallization	80–180
Textiles, leather, and laundry services	Dyeing baths, drying, finishing	60–160
Wood products and prefabricated building products	Lumber kiln drying, panel pressing	60–200
Light manufacturing and assembly	Paint cure, parts washing, fabrication-hall conditioning	50–180
Mining and oil and gas	Produced-water heating, glycol regeneration, remote-site building heat, mid-stream gas processing	60–200

Aggregate energy demand in the food and beverage subsector in Canada is on the order of 100 PJ/yr, predominantly supplied as low-pressure natural-gas steam (Natural Resources Canada 2024). The pulp and paper subsector is among the largest industrial consumers of natural gas in Canada and the largest non-electric thermal load on a per-establishment basis (Natural Resources Canada 2005). ECCC's NIR attributes the majority of stationary combustion emissions in Canadian manufacturing to natural gas, with secondary contributions from heavy fuel oil (concentrated in Atlantic Canada) and refinery still gas; aggregate sectoral activity used to disaggregate these emissions in CIMS is taken from the CEUD and the NIR (Environment and Climate Change Canada 2024). The U.S. EIA's MECS provides a parallel accounting for U.S. industry: conventional boilers and process heaters account for over 70 % of manufacturing fuel use (U.S. Energy Information Administration 2021).

6.3 Technology Set for Low- and Zero-Emission Supply

This section enumerates the low- and zero-emission supply technologies admitted to the low-temperature service nodes in the CIMS industrial sub-models, together with their assumed performance parameters and data sources. Technology efficiencies and coefficients of performance (COPs) used to parameterize the technology competitions are summarized in Table 6.2. Combustion boiler efficiencies are estimated based on data from MECS; heat pump COPs are based on inputs from IEA Heat Pumping Technologies Programme Annex 58 (U.S. Energy Information Administration 2021; IEA Heat Pumping Technologies Programme 2023).

Table 6.2: Assumed efficiencies and coefficients of performance for low-temperature heat supply technologies

Technology	Efficiency / COP	Maximum sink temperature
Gas boiler (reference)	80–85 %	250 °C
Liquid fuel boiler	80–85 %	250 °C
Coal boiler	85–90 %	250 °C
Biomass boiler	75 %	250 °C
Electric resistance / electrode boiler	99 %	250 °C
Heat pump, sink 100 °C	COP 4.0–6.0	100 °C
Heat pump, sink 100–150 °C	COP 2.5–4.0	150 °C
Heat pump, sink 150–200 °C	COP 2.0–3.0	200 °C (demonstration)
Mechanical vapour recompression	COP 5–10	Evaporator duty only
Solar thermal, flat-plate / evacuated-tube	n/a	100 °C
Solar thermal, linear concentrating	n/a	200 °C

Combustion boilers and electric resistance boilers are assumed to have a model life of 25 years; industrial heat pumps and mechanical vapour recompression units are assumed to have a model life of 15 years; solar thermal collectors are assumed to have a model life of 20 years. Both existing and new boilers are limited to annual capacity factor thresholds that vary by industrial sub-model and are calibrated to historic data. All values are stated in 2020 CAD terms; capital cost and fixed operating-and-maintenance parameters are populated separately in the techno-economic data files referenced by the relevant industrial sub-model.

6.3.1 Industrial Heat Pumps

Closed-cycle vapour-compression heat pumps upgrade ambient or recovered low-grade heat to a useful sink temperature. Commercially available products reach sink temperatures of approximately 165 °C; units operating at 200 °C with novel working fluids are at the demonstration stage (Arpagaus et al. 2018; Zühlsdorf et al. 2019). On the Canadian electricity system, with weighted-average grid emissions intensity well below that of natural gas combustion, replacement of gas-fired boilers with heat pumps is estimated to reduce direct CO_2 emissions by 70–95 % (IEA Heat Pumping Technologies Programme 2023). Mechanical vapour recompression, an open-cycle heat pump configuration applied to evaporators in dairy, sugar, and pulp operations, is the most mature variant and is represented as a distinct technology in the food and beverage and pulp and paper sub-models.

6.3.2 Direct Electrification

Electric resistance and electrode boilers deliver low-pressure steam at near-unity end-use efficiency at lower capital cost than heat pumps, and are represented as the simplest like-for-like replacement for packaged gas-fired boilers. Application is favoured in provinces with low-emissions grids — Quebec, British Columbia, Manitoba, and Newfoundland and Labrador, where average grid emissions intensity is below 30 g CO_2/kWh — and in retrofit configurations where capital constraints favour low first-cost equipment (Environment and Climate Change Canada 2024). Direct-contact electric processes (induction, infrared, microwave, and dielectric heating) are commercially deployed for specific drying, curing, and surface-heating duties and are admitted as candidate technologies at the corresponding service nodes (Madeddu et al. 2020).

6.3.3 Solar Thermal

Flat-plate and evacuated-tube collectors economically supply heat up to approximately 100 °C; linear concentrating systems (parabolic-trough and linear-Fresnel) extend the addressable range to 200 °C and beyond. Schoeneberger et al. estimate that approximately half of global industrial heat demand falls within the temperature window addressable by current commercial solar thermal technologies (Schoeneberger et al. 2020). Within Canada, the resource is concentrated in the southern Prairies and the southern interior of British Columbia; representation in CIMS is restricted to the relevant provincial sub-models, with capacity factors calibrated to regional direct normal irradiance (International Energy Agency 2024).

6.3.4 Bioenergy

Biomass-fired boilers, anaerobic-digestion biogas, and renewable natural gas from landfill and agricultural sources are admitted as drop-in low-carbon supply technologies in facilities configured around combustion equipment. Biomass is the dominant on-site thermal energy source in the Canadian pulp and paper subsector and is also relevant in food processing operations co-located with feedstock supply (Natural Resources Canada, CanmetENERGY 2026). Outside the pulp and paper subsector, deployed bioheat in Canada is concentrated at the 50 kW–5 MW thermal capacity range tracked by CanmetENERGY's Canadian Bioheat Database; the 2023 survey update reports that more than 90 % of catalogued commercial, institutional, and small-industrial systems use either clean wood chips or energy chips (approximately 45 %, dominant in small-industrial and agricultural installations) or wood pellets (approximately 46 %, dominant in commercial and multi-unit residential applications), with hog fuel and agricultural residues uncommon at this scale and concentrated in pulp and paper and adjacent forest-products operations (Natural Resources Canada, CanmetENERGY 2023). Regional fuel-type preferences are pronounced: wood chips dominate in Quebec and British Columbia, while pellets account for nearly all installations in New Brunswick, Ontario, and the Northwest Territories. These distributional facts inform the technology entries admitted to the bioenergy supply node feeding the Light Industrial sub-model, and the regional weighting applied to wood-chip versus pellet capital and operating cost assumptions. Sustainable supply is the binding constraint at the system level and is represented in CIMS through fuel supply curves rather than at the device level.

6.3.5 Geothermal and Recovered Waste Heat

Deep geothermal reaches useful process temperatures in parts of western Canada; shallow geothermal coupled with a heat pump is deployable across all provinces for facility heat. Independent of geothermal extraction, every industrial site is itself a source of low-grade waste heat — compressors, condensers, flue gases, and effluents — that is conventionally rejected to the environment. National-laboratory analyses estimate that 20–50 % of industrial process heat could in principle be supplied via recovered waste heat (Thekdi and Nimbalkar 2014). Process integration (pinch analysis) and heat-pump-assisted recovery are represented in CIMS as a service-side reduction in the demand directed to other supply technologies, parameterized by sub-model based on recoverable-heat fractions reported in MECS and CanmetENERGY studies, rather than as a competing supply technology at the service node.

6.4 Cross-Cutting Treatment

6.4.1 Thermal Energy Storage

Pressurized hot water, molten salt, and solid-media storage shift electrified heat loads to off-peak hours and extend solar thermal coverage to continuous-process duty cycles. Storage is treated in CIMS as a distinct technology that competes at the same service node as the heat-supply technologies it enables, with round-trip efficiency, capacity, and discharge-duration parameters drawn from IEA reference values (International Energy Agency 2022).

6.4.2 Hydrogen

Direct combustion of hydrogen below 200 °C is not competitive with electrification under most modeled conditions: round-trip energy efficiency from electricity to hydrogen to heat is substantially lower than direct or heat-pump electrification, and the system value of hydrogen is concentrated in genuinely high-temperature or feedstock applications. Hydrogen is therefore not included in the default low-temperature technology set; it remains available in the broader CIMS technology library for the medium- and high-temperature sub-models (Thiel and Stark 2021; Rissman et al. 2020).

6.4.3 Endogenous and Exogenous Parameters

Provincial grid emissions intensity, electricity price, and natural gas price are determined endogenously by the CIMS energy supply sub-models. Sectoral output projections, technology capital and fixed operating-and-maintenance costs, model lifetimes, and biomass supply curves are exogenously specified. The model includes the option to accelerate the rate of efficiency improvement at additional cost in response to rising energy prices, consistent with the treatment of efficiency in other CIMS industrial sub-models.

6.5 Implications for the CIMS Industrial Sub-Models

The technology competitions in the food and beverage, pulp and paper, chemicals, mining, and light manufacturing sub-models admit, at the relevant service nodes, conventional gas, oil, and (where regionally relevant) heavy fuel oil boilers, used for calibration to CEUD and NIR baseline data; electric resistance and electrode boilers, parameterized by provincial industrial electricity rates; industrial heat pumps disaggregated by sink-temperature class (100 °C, 100–150 °C, 150–200 °C) with explicit COP and source-temperature assumptions drawn from IEA HPT Annex 58; mechanical vapour recompression where evaporation duties exist; biomass and biogas boilers, with fuel availability governed by the bioenergy supply sub-model; and solar thermal in southern Prairie and southern British Columbia provincial sub-models, parameterized to regional irradiance (Natural Resources Canada 2024; Environment and Climate Change Canada 2024; IEA Heat Pumping Technologies Programme 2023; Natural Resources Canada, CanmetENERGY 2026; Schoeneberger et al. 2020).

Existing boiler stock, distinguished by sub-model and temperature class, is estimated based on CEUD and NIR base-year data. All existing boiler stock is assumed to stay online for modeled lifetimes past the base year, after which replacement is required. Policy or other drivers may drive earlier retirement and replacement of some of these existing units. Capital costs, retrofit feasibility (as defined in the technology chapter), and electricity-price sensitivity vary substantially across these technologies. The lifecycle-cost formulation already in CIMS is applicable without modification once the techno-economic parameters are populated. The present chapter is intended to provide the technical basis for those parameter selections.

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Part III

results

7 Policy Costing & Outputs

7.0.1 Policy costing methods associated with ISTUM and early CIMS

Issues associated with deriving costs from its outputs guided much of CIMS' development. The cost methodologies in the following sections are described in brief and outline some of the issues in estimating costs.

The ISTUM model was initially designed for energy policy analysis, mainly for assessing how fuel consumption and emissions would change with a given energy policy. The emphasis was on marginal costs, not total or absolute expenditures. The first consistent costing concept was the techno-economic cost, or TEC.

TODO: Should we have standard reporting for costs below?

Techno-economic Cost (TEC)

TEC is the difference in estimated (ex-ante) financial expenditure on capital, labour, and energy between the Business-as-Usual and Policy cases. It does not usually (unless otherwise specified) contain carbon taxes as these are considered a transfer internal to society. It is possible to get negative TEC costs, or benefits, especially in transportation where increased use of transit and walking and cycling reduces financial costs, implying that some actions induced by policy may be profitable. There are significant complications as well – do those who switch to transit and walking and cycling continue to own and pay for cars? This reflects the challenging question about how vehicle acquisition will be affected by policies to reduce vehicle use. TEC for all consuming sectors are reported with and without electricity price increases, which are sub-regional transfers that balance out to zero at the regional level once the financial effects on electricity producers' revenues are included.

The ISTUM sub-models were integrated under the energy demand and price feedback system in the mid-1999 to 2000 period, thus creating the first version of CIMS. During this period, the model was used for a series of large GHG abatement analyses for Canada. The thrust of these analyses was to integrate and assess a series of emissions reduction actions proposed by panels of interested parties, the Issue Tables of Canada's Climate Change National Assessment Process. The most effective actions were determined by minimizing their welfare costs, as opposed to their pure financial costs. This led to the development of the Perceived Private Cost methodology (Bataille et al. 2002, Laurin et al. 2003, MKJA/EMRG 2000).

Perceived Private Costs (PPC)

PPC are calculated as the GHG tax multiplied by the marginal amount of GHG reductions that occurred at each GHG tax level. The area under the resulting GHG abatement cost curve represents the perceived cost of the actions by the GHG charge. The PPC methodology estimates welfare losses that represent in part the unwillingness of all consumers to switch to technologies that reduce emissions. This fostered development of the Expected Resource Cost methodology, which attempts to more realistically depict the costs and risks associated with a given GHG reduction policy.

Expected Resource Costs (ERC)

ERC is PPC that has been adjusted for risk, option value, and general inefficiency. It is equal to TEC+(PPC-TEC)*0.75. The difference between TEC and PPC that is allowed as ERC is a "rule-of-thumb" estimate for what percentage of the difference can be considered real equipment failure risk, as opposed to inefficient resistance to price signals. It represents that perceived cost, which could turn into real financial cost. Negative ERC is possible, especially at lower GHG charges, but not likely (Bataille et al. 2002, Laurin et al, 2003, MKJA/EMRG, 2000).

Table 10 summarizes the three early costing methods used for CIMS.

Table 8. Types of Costs Used in Early Incarnation of CIMS

Distribution of Costs Under Electricity and Refining Demand Changes

Electricity demand changes in all sectors in response to increased electricity prices, leading to an issue of how to distribute GHG reductions as well as TEC and PPC charges. GHG reductions from reduced electricity consumption were accorded to the direct consumer (based on the policy GHG intensity of electricity), who faced increased electricity prices, while electricity producers were accorded the remaining GHG intensity reductions, the TEC costs generated by their intensity reductions plus their changes in revenue. These net transfers, while positive at the sub-regional level, disappear at the regional level.

The Relationship Between TEC, ERC, Option Value and Welfare Costs

The cost difference between ERC and TEC estimated during early analyses was described as risk, or "failure risk", which is related to the concept of option-value (Pindyck 1991, Dixit 1992, Dixit and Pindyck 1994). Option value is the expected gain from delaying or avoiding an investment while waiting for new information that might lead to a better decision. Thus, ERC are anticipated ex-post costs, and it is calculated as the sum of expected financial costs (TEC), lost option value, consumers' surplus losses internal to residential and transportation (internal shifts away from business-as-usual (BAU) preferences within the sub-models) and finally, when demand feedbacks are included, consumers' surplus losses related to demand adjustments for domestic consumers of domestic goods. It may therefore be considered a measure of personal welfare costs, not including negative environmental externalities (e.g. poorer quality air) or positive public goods (e.g. education, defence or public parks).

7.0.2 Policy Costs Associated With Changes in Demand

The inclusion of final demand feedbacks required the addition of five new policy-costing concepts. In all these concepts, financial output is defined as a sector's sales revenue, which includes both value-added and purchased inputs. Value-added is net profits plus the cost of all expenditures on land (unprocessed "gifts of nature"), labour and capital (Lipsey et al. 1988).

- The cost of production effect on sector financial output, which is the increased financial expenditures required to adapt to the policy, or TEC.
- The demand effect on sector financial output, which is the reduction in production expenditures associated with lower output.
- The absolute change in sector financial output, which is the sum of the cost of production and demand effects.
- Consumers' surplus losses associated with demand changes in transportation and residences.
- Changes in value-added associated with cost of production and demand changes in manufacturing.

7.0.2.1 The Cost of Production and Demand Effects on Sector Financial Output

The cost of production effect captures the internal capital investment and fuel choice adjustments of the model, or the TEC costs endogenous to the model. It is the anticipated financial cost of technical adjustment to the policy. The demand effect captures the effects of the demand feedbacks system, or reduction in output, as measured by expenditure compared to the business-as-usual case (BAU). The third is the sum of the first two. These three concepts capture a portion of the anticipated financial cost of a policy. They do not include the cost of buying emission permits from foreign sources or the cost of imports that replace domestically produced goods for domestic consumption. The fourth item, consumers' surplus losses, represents lost consumer welfare associated with reduced residential expenditure and mobility, while the fifth represents the lost value-added to manufacturing firms due to policy.

Most GHG policies will increase the cost of production for a given service, increase its price, and thus reduce its demand. This increase in the cost of production will also include the value of carbon charges that are imposed on GHG emissions. These charges are transfers and as such must be subtracted from regional and national costs. Additionally, if demand is inelastic (unresponsive to price) for a given service, the dollar value of the sector, as valued by cost of production, may increase faster than demand reductions, leading to an overall increase in sector value to the economy. If demand is elastic, the opposite may occur.

The standard outputs from the sub-models include BAU and policy values for new physical capital investment associated with consuming energy, labour associated with the latter capital, and energy costs, as well as changes in physical output. It also produces BAU and policy costs per unit output, including GHG charges. These GHG charges are isolated by calculating policy emissions multiplied by the GHG price. When the micro and macro economic dynamics are turned on, part of the difference between the BAU and policy cases may be physical output changes while the rest will be cost of production differences due to internal stock adjustments. CIMS only captures a portion of the price that the consumer sees; both the BAU and Policy cost per unit are adjusted accordingly. The cost of production and demand effects on sector financial output (SFO) (Equations 23 & 24), and the absolute change in financial expenditure are calculated as follows (Equation 25).

(Equation 28)

(Equation 29)

(Equation 30)

Where:

QGHG = GHG emissions (Tonnes CO2e)

PGHG = GHG price ($/Tonne CO2e)

Qpolicy = Policy physical output of good or service X

QBAU = BAU physical output of good or service X

Ppolicy = Policy cost per unit of good or service X

PBAU = BAU cost per unit of good or service X

The absolute change in sector financial output could be positive or negative depending on the elasticity of demand – if the cost of production rises faster than demand falls, then the absolute change in financial output could be positive. This result is common with less aggressive policies. The demand effect on sector financial output will always be negative for normal goods, whose demand fall as their price rises. The cost of production effect on sector financial output is usually positive, except for the case where there are decreasing returns to scale.

7.0.2.2 Consumers' Surplus Adjustments in Residential and Transportation

Reductions in residential activity and mobility reflect consumers' surplus losses. Consumers' surplus is the area between the demand curve, which symbolizes the marginal willingness to pay for a service or good, and the market-clearing price of the good or service (Figure 8).

Figure 4. Consumers' Surplus Diagram

The consumers' surplus loss between BAU and Policy is then calculated as BAU minus the policy. When calculating the direct welfare costs of a policy, consumers' surplus loss is added to the ERC value already calculated. Table 11 summarizes the financial and welfare costs adjustments made to the various sectors when their output changes.

Table 9. Financial, Welfare Cost and Value-added Adjustments Induced by Changes in Output

The lack of an output adjustment for welfare costs or value-added for the commercial sector is partly due to the nature of the Commercial and Institutional sub-model. It is composed mainly of building shells, HVAC, miscellaneous electricity and air-conditioning systems for government and office buildings, retail space, hospitals, schools, etc. CIMS does not model what these spaces are used for, and thus cannot account for how financial output or value-added may have changed.

7.0.3 Outputs from CIMS

Table 12 provides the outputs specific to the sub-models (i.e., for when they are run by themselves without feedbacks), while Table 13 provides the extra outputs necessary for working with the CIMS' micro and macro economic systems of dynamics (also referred to as feedback systems).

Table 10. Sub-model Outputs Existing Prior to This Research

Sub-models Annualized lifecycle costs by sector by year by technology based on behavioural discount rates (20-80%) (See Table 5).

Fuel use and emissions by sector by year by technology.

Service and auxiliary use by sector by year by technology.

Total stock use by sector by year by technology.

New stock use by sector by year by technology.

Retrofits by sector by year by technology.

Technology cost changes due to declining capital cost function

Table 11. Supply and Demand Feedback Outputs Added as a Result of This Research

Supply and Demand feedbacks Annualized lifecycle cost per unit produced for all sectors by year based on financial cost of capital for both the BAU and Policy case; the model operator sets the cost of capital (e.g. 7.5 or 10%)

BAU and Policy final demand for each sector's product or service by year.

BAU and Policy disaggregated energy demand for all sectors by year.

Changes in disaggregated capital, operations and maintenance, and energy costs by sector by year.

Changes in GHGs and selected criteria air contaminants (CACs), energy and service use by technology, sorted by technology competition, for BAU and Policy by sector by year.

Changes in net exports of key energy commodities from BAU to Policy

Changes in energy prices by year

Changes in the financial cost of production split into the cost of production and demand effects for industry and manufacturing

Changes consumers' surplus for personal transportation and residences

Changes in value-added for industry and manufacturing

8 Outputs

TODO: Results viewer

- Need to revise how 'short path' is created once markets and/or other countries are added
- Populate Tech Details only from pivot data. Need to add:
  - fom
  - fic
  - dic (need to calc in model)
  - ucc (need to calc in model)
  - recycled revenue (at techs?)
- Organise tree to follow model description order
- Auto generate tree? Need hierarchy diagrams even if not auto generated.
- Electricity sector -> grid intensity

Three main classes of output are created by the model:

1) Sub-sector specific reports (DRAFTED)
   a) fuel and emissions for a given sector (FUELA) for BAU and Policy, (B or C-Region-FUELA, e.g. BONFUELA.
   b) technology levelized costs for a given sector (LEVEL) – same pattern as FUELA.
   c) new and total technology market share reports for a given sector (NEWMA and TOTAL) – same pattern as FUELA.
   d) technology node competition reports for a given sector (REGION-NC-YEAR, e.g. "ABNC2005"
2) Semi-aggregate reports (TO BE DONE)
   a) changes in physical output, sorted by sector and by region (CCOMP)
   b) changes in sector capital, fuel, emissions and labour costs, sorted by sector and by region (CIMCOSTA & CIMCOSTB)

    c) changes in sector energy use (by market class Elec., NG, coal and RPP) and GHG emissions, sorted by sector and by region (DCOMP)
    d) changes in sector cost per unit (inclusive of emissions charges), sorted by sector and by region (LCOMP)

3) Aggregate report (TO BE DONE)
    a) CIMS Outputs (As of March 2007)

8.1 Sub-sector Reports

Four report types are available. In each type of report, the analyst opens a spreadsheet where data can be manipulated using spreadsheet functions. The report can then be reformatted, printed, or saved under another name.

8.1.1 Fuel and Emissions Report

Data concerning fuel consumption and emission generation are saved for each year of the simulation. The spreadsheet entitled B(BAU) or C(Policy) Region (e.g. BC) FUEL, e.g. CBCFUELA, receives these data. The model's report generator produces a table containing: (1) a list of all fuels and emissions, (3) consumption/production of those fuels and emissions for each year, and (4) a summation of fuels consumed and emissions output. Note that, if fuel/emission units are not uniform, this summation will be erroneous. Figure 3.1 displays the layout of a typical aggregate fuel report.

Figure 3.1 ISTUM's Fuel Report

Typically, technologies are grouped into Aggregate, Major processes, Service categories and Product type categories.

Aggregate reports include the cumulative energy consumption of all the technologies in the industrial branch under analysis and are generated for every industry. Note that this is the sum of consumed and produced energy. To determine the amount of cogenerated or self-generated energy, the analyst should refer to the NC files which show al energy inputs and outputs by technology competition. No technology can contribute to the cumulative consumption/production of more than one group in each category.

Figure 3.1 ISTUM's Fuel Report

8.1.2 Total Stock Report

The table generated in the TOTAL spreadsheets contains data on the sum of operating old and new stock inventories for that year. The stock is listed in output units for that technology. As with the other report types, the table's header contains specific information on the scenario and economic criteria applied to produce that output. To the right of the total stock table, one finds a table of the share of total stock for competing technologies (in percent) that each technology has obtained. Figure 3.23 displays the layout of a typical Total Stock Report.

The lower portion of this table provides information concerning changes in stock consequent to retrofitting. Each set of lead and retrofit technologies show the lead technology first followed by the technology(ies) that can be retrofit to it. The data indicate the number of units not retrofit (the value associated with parent technologies) and those that were retrofit (the values associated with the retrofit technologies).

Figure 3.2 ISTUM's Total Stock Report

8.1.3 New Stock Report

As in the Total Stock Report, the two tables generated by the New Stock Report function of CIMS, show new stock additions in each year of the simulation and the market share of that stock purchased in that simulation year. This NEWMAR spreadsheet header, like the others, displays specific information concerning scenario and economic criteria. Figure 3.3 displays the layout of a typical New Stock Report.

Figure 3.3 ISTUM's New Stock Report

8.1.4 Levelized Costs Report

The LEVEL report lists each technology and contains tables displaying two categories of information:

1) The first set of columns include data used to generate the levelized costs. The first numeric column displays capital costs (CC) of each technology, followed by its annualized CC (CC x CC recovery factor, found in the sixth column). The third column shows the sum of annualized CC and costs of operations and maintenance (excluding fuel and emissions costs) for each technology. This, divided by the material output found in column 4 generates the lifecycle cost per unit output in column 5. Column six indicates whether any non-cost parameter was applied to the capital costs (CC x non-cost parameter).

2) The second set of columns, each headed by a simulation year, presents the levelized cost of that technology in that year. The levelized cost is the sum of the per-unit cost (column four) and the annual fuel (and emission) costs. (Figure 3.4)

The lower portion of this table provides information concerning retrofits applied in that simulation. Each set of lead and retrofit technologies, displayed in separate groups, show the lead technology first followed by the technology(ies) that can be retrofit to it. Note that the capital cost of the lead technology is always 0 since it is considered a sunk cost. The capital costs associated with the retrofit technologies represents the cost of retrofit purchase and installation.

Figure 3.4 ISTUM's Levelized Cost Report

8.1.5 Node Competition Report

There is a separate node competition report for every sector in every region in every year. They are title (region)NC(year)J, e.g. for all sectors in Alberta the 2005 report is called "ABNC2005J.xls". The region codes are: BC, AB, SK, MB, ON, PQ, ATL (for Atlantic).

The node competition report presents the results of a run as a comparison between BAU and POLICY in terms of fuel use, services and emissions. The key difference between this report and the last four is that it organized by technology competition. As there a several fuels, services and emissions in CIMS, these reports tend to be very large. Figure 6 on the next page provides a sample node competition report for the baseload electricity production competition in Alberta in 2005. The shown figure gives the difference in stock, fuel use and emissions, while further to the right on the spreadsheet will be the BAU and policy absolute values. Stock is valued in the output unit of the sector in question, energy in GJ, and emissions in tonnes.

Figure 6: Sample node competition report – base load electricity production Alberta 2005

8.2 End user reports

TODO: Reports of energy use, financials, intensities, travel, etc. at end user level

8.3 For example, household energy spending, average travel and costsMACC Curve

TODO: EDF report, MACC 2.0

8.4 Kaya Decomposition

TODO: Decomposition of results (to also use in policy attribution)

Include automatic waterfall chart?

Factors:

- Activity
- Etot / Activity (efficiency)
- Eff / Etot (fuel switch)
- GHGtot / Eff (fuel decarbonisation)
- GHGnet / GHGtot (CCS?)

8.5 Attributing emissions reductions to policies

TODO: Shapley values using SHAP, LIME, or QII